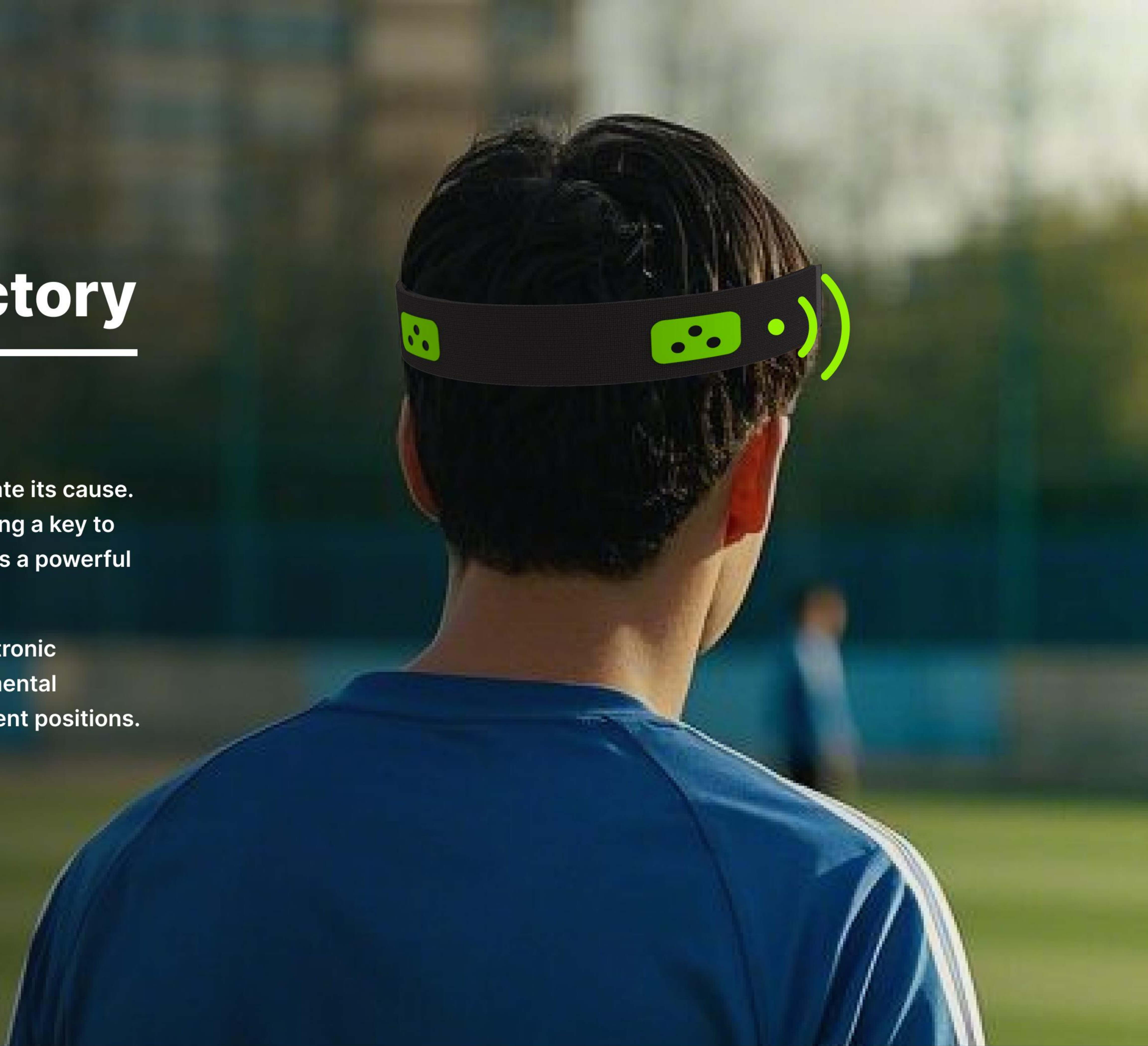


Simulation- Key to Victory

- Interaction Design
- Product Design
- User experience

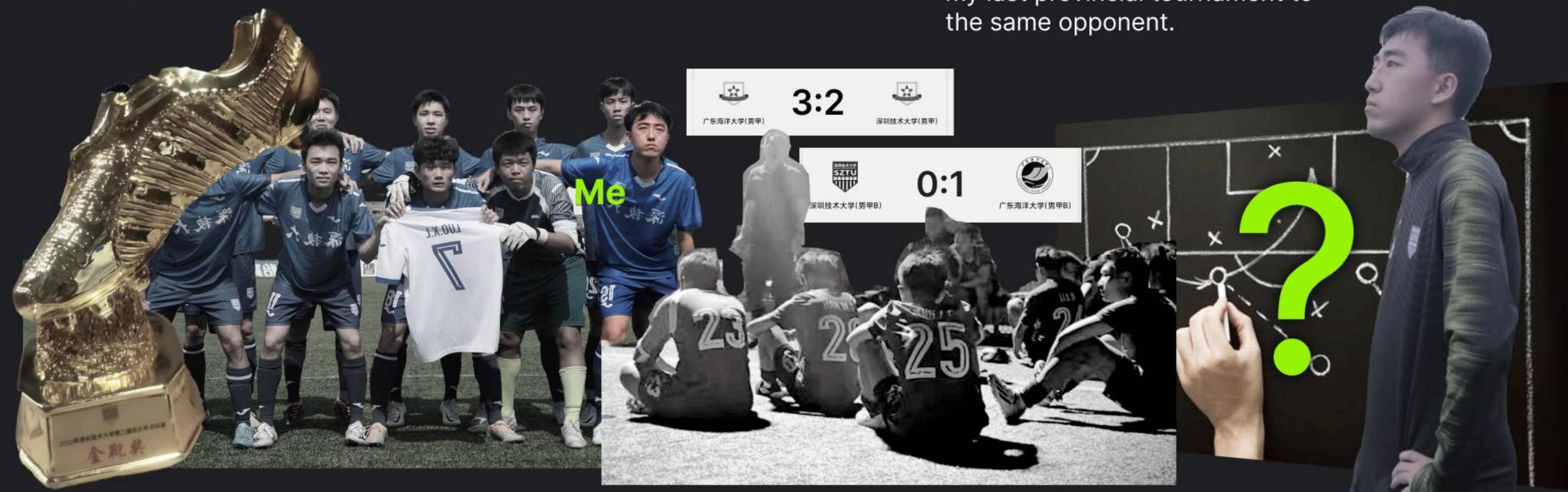
A key loss during my school team career drove me to investigate its cause. Surprisingly, our opponents trained less than we did, suggesting a key to improving efficiency. Interviews revealed simulation training as a powerful tool, inspiring me to analyze its elements and innovate.

Using Harris chart analysis, I designed a tiered-response electronic headband that alerts players to on-field scenarios, boosting mental resilience and reaction speed while tailoring training for different positions.



Inspiration

- I In my freshman year, winning the Golden Boot award in the freshman game, I made the varsity team.
- II I have participated in various competitions at provincial and municipal levels with the school team.
- III A couple of heartbreaking losses in key tournaments, particularly my last provincial tournament to the same opponent.
- IV I'd like to know what exactly causes people to underperform in key games.



- My group is a general student group (no professional training experience)
- There's not much difference in the level of either the players or the coaches.
- I think they may have had more practice time together and so performed better.
- Surprisingly I've found a couple of strong teams that train less than we do.

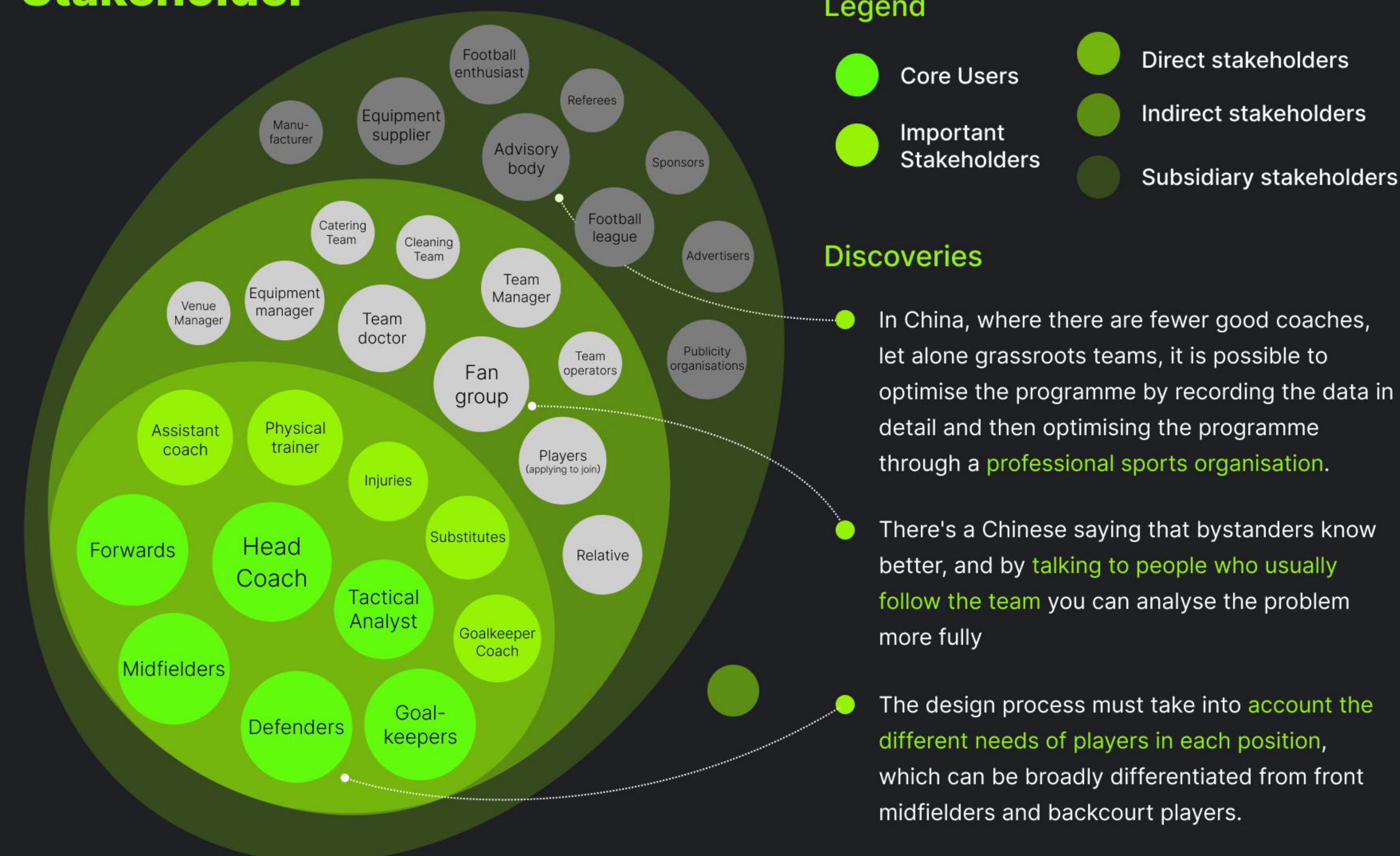


What are the key factors that balance the scales?

Research

Primary research is primary, secondary research is supportive

Stakeholder



Interviews for people in different roles

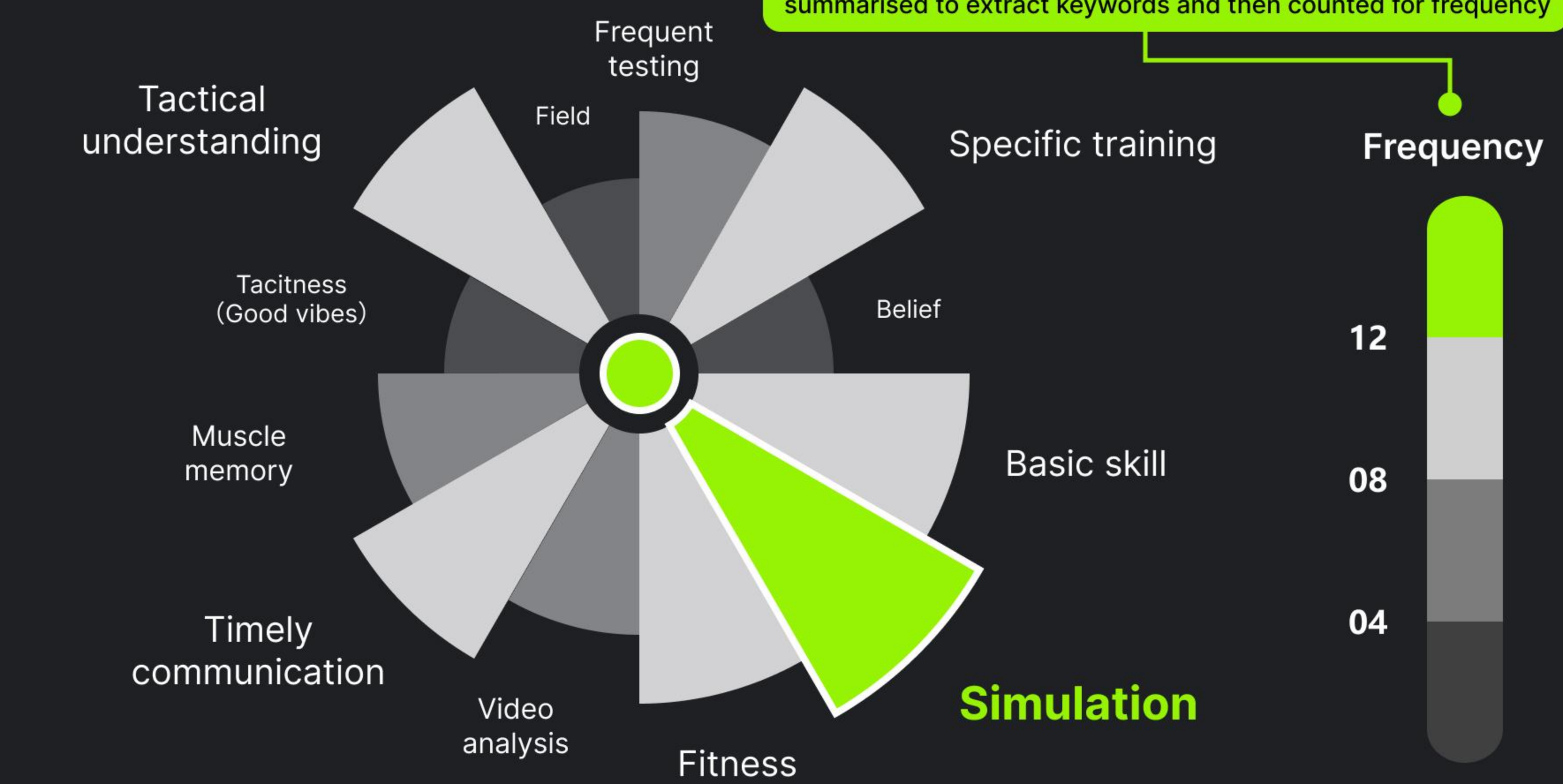
User Interviews

- What do you think is the problem with the current team's training inefficiencies?

Which training do you feel has been effective in improving your performance during a race?
- For players and coaches

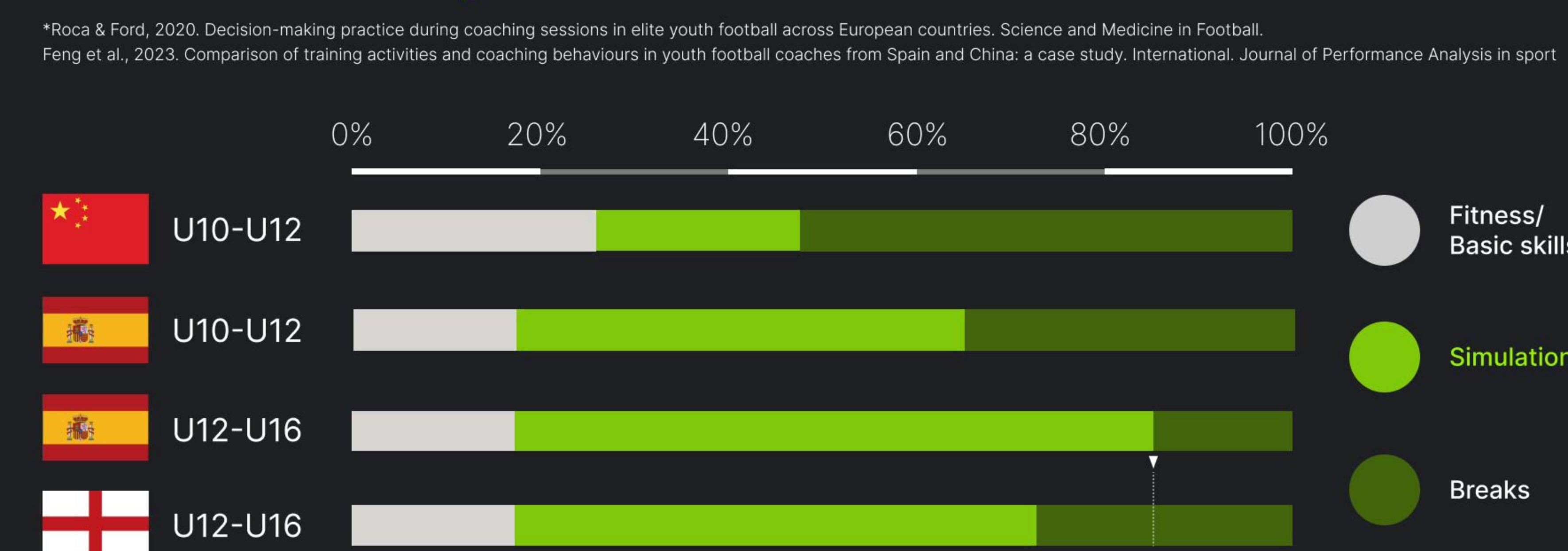
- For fans and staff member
- Through observation and knowledge, which part of the training do you think is the most important?

Results statistics



Information analysis

*The structure of elite youth football training sessions



The comparison reveals the importance football powerhouses place on simulation training, which to some extent supports our interview findings.

These two teams just happened to be the opposing sides in the Euro 24 final, so perhaps a little difference in simulation training as a teenager played a bit of a role

Elemental analysis-simulation

Match scenarios

The desire for championships and victories fuelled the players

Give it all you've got. After all, no one wants to lose

High-intensity environments are more likely to be out of control

Your opponent will never give you time to think

The atmosphere built by the fans is exhilarating

For example, the Spanish player Reina would fill up his car with petrol before every match

Players need to adapt to different venues and facilities

Coaches and referees will show a different style than they do in practice

Aims Pursued

Range of motion

Emotional Condition

Decision-making Time

Fan Vibe

Personal Superstition

Football field facilities

Coaches & Referees

Just do what's on the training schedule, lack of enthusiasm

No need to fight hard to avoid hurting teammates

The atmosphere is much more relaxed and there are few emotional outbursts

Players have time to think about their next move

Only team officials and family visit the training ground

Superstition exists but it's not the same as an official game

Familiarity with the training ground, facilities, and equipment

Coaching is more patient and the role of referee is difficult to revert to

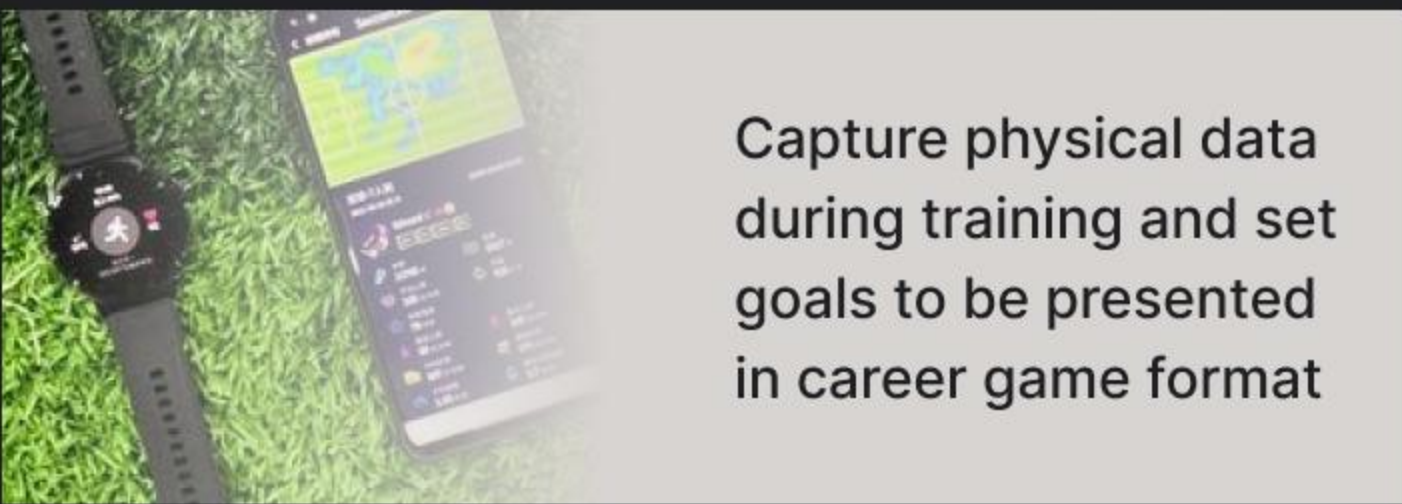
Initial programme

- 1

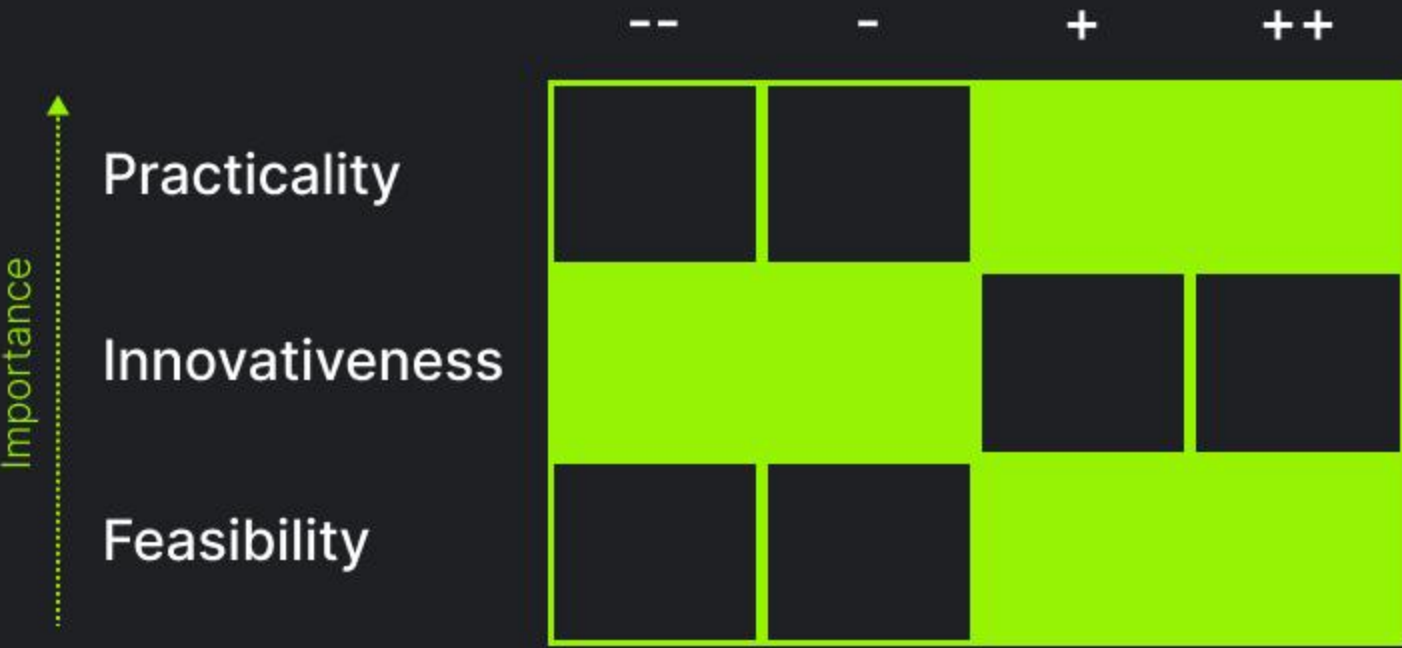
Remove the more subjective elements of 'range of motion' and 'personal superstitions' and extend the programme with the other elements.
- 2

Potential scenarios were analysed using the Harris profile method and the highest scoring scenarios were developed and designed.

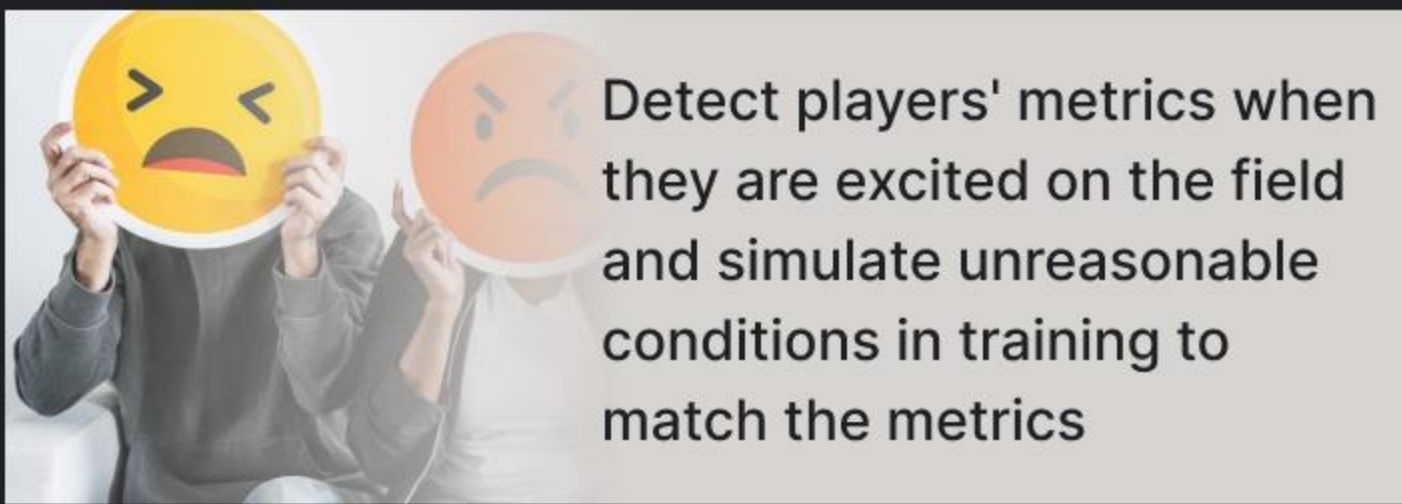
Aims Pursued



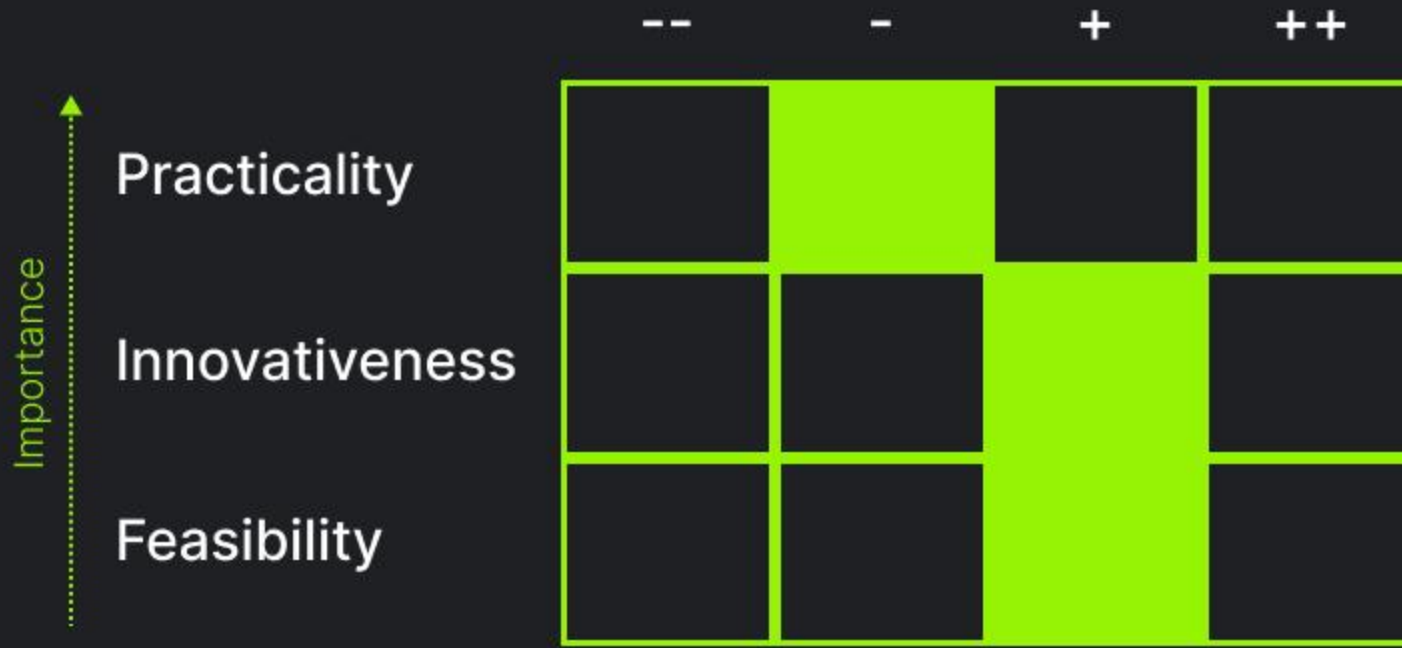
Capture physical data during training and set goals to be presented in career game format



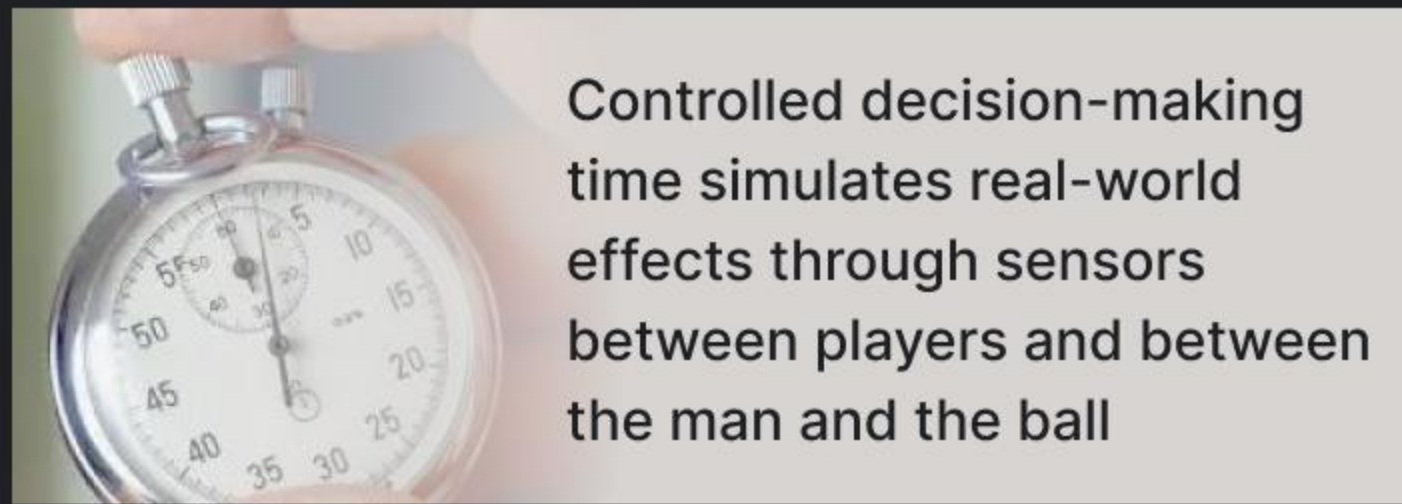
Emotional Condition



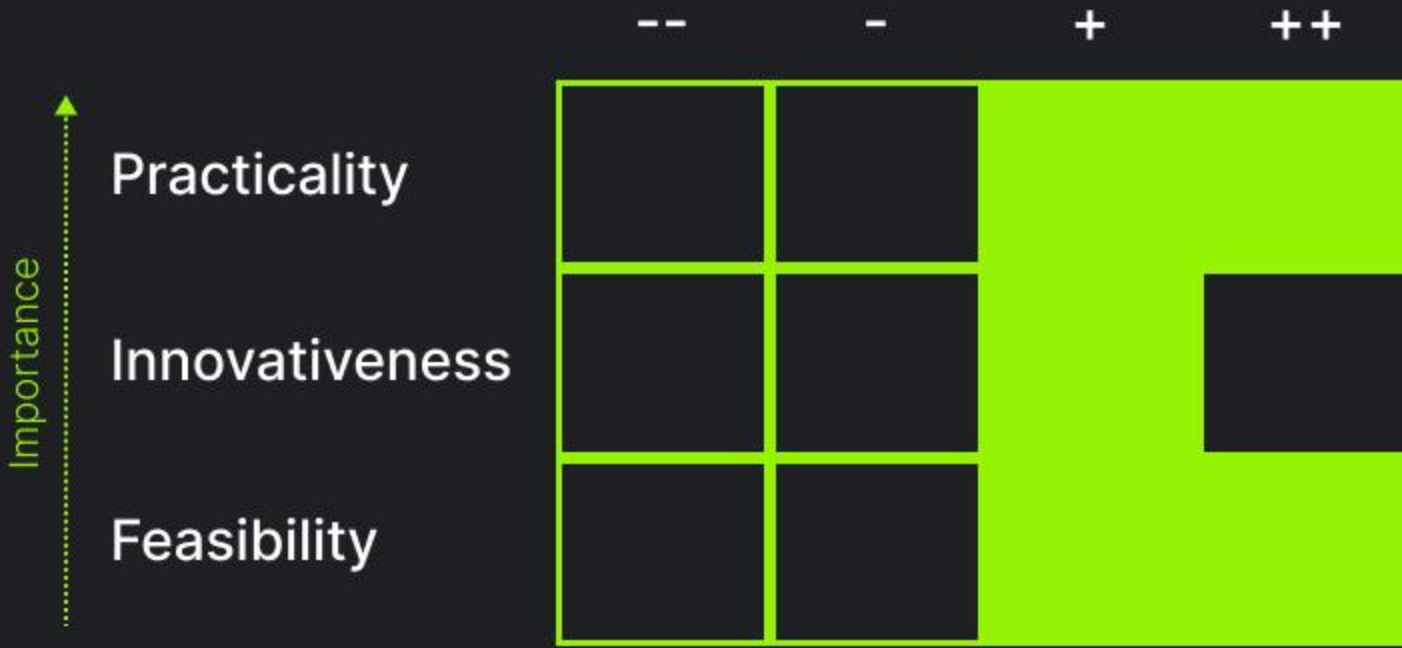
Detect players' metrics when they are excited on the field and simulate unreasonable conditions in training to match the metrics



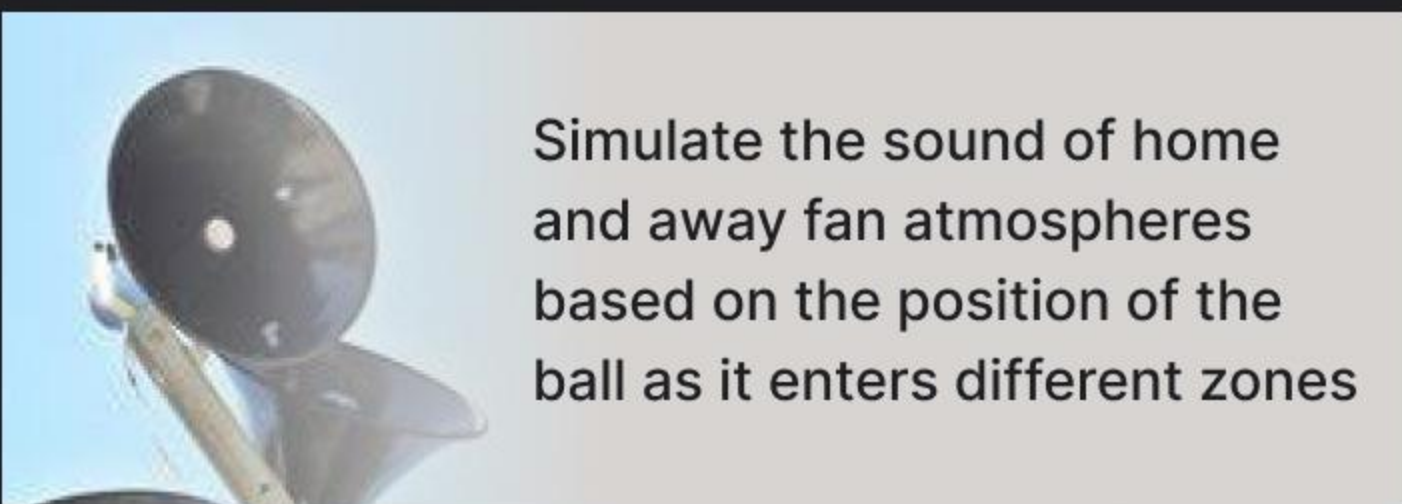
Decision-making time



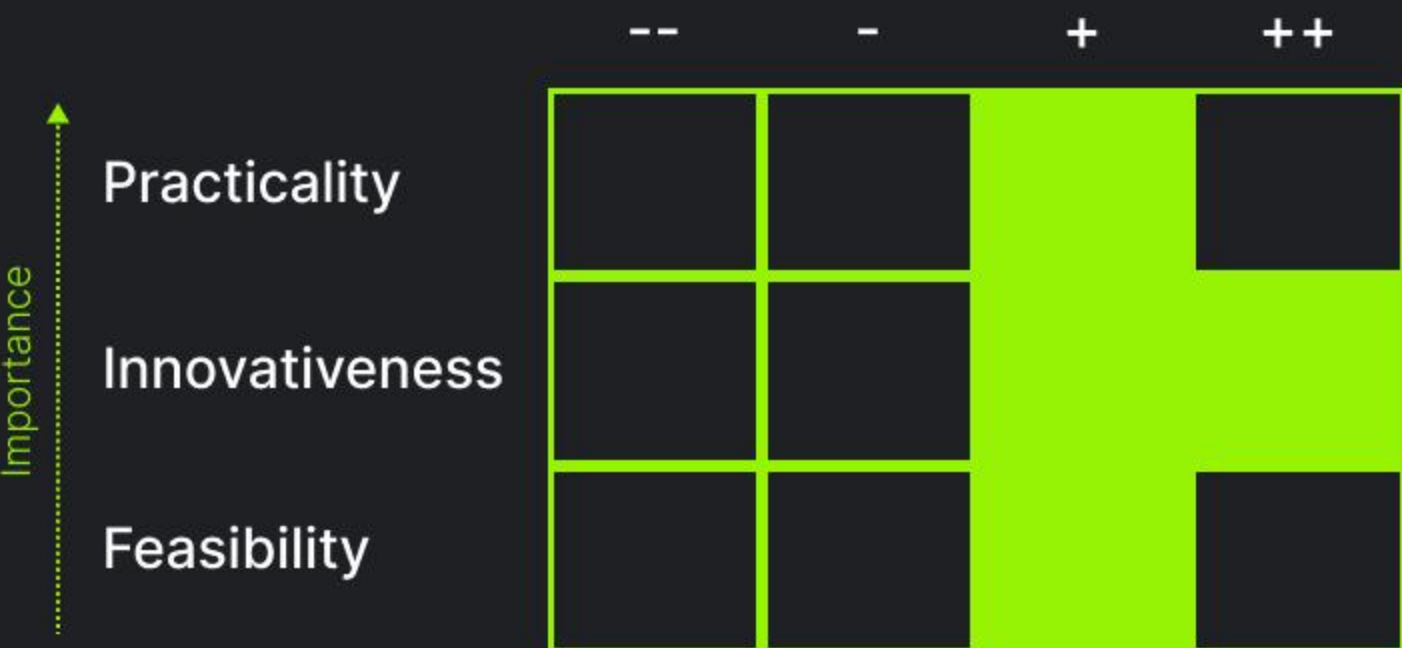
Controlled decision-making time simulates real-world effects through sensors between players and between the man and the ball



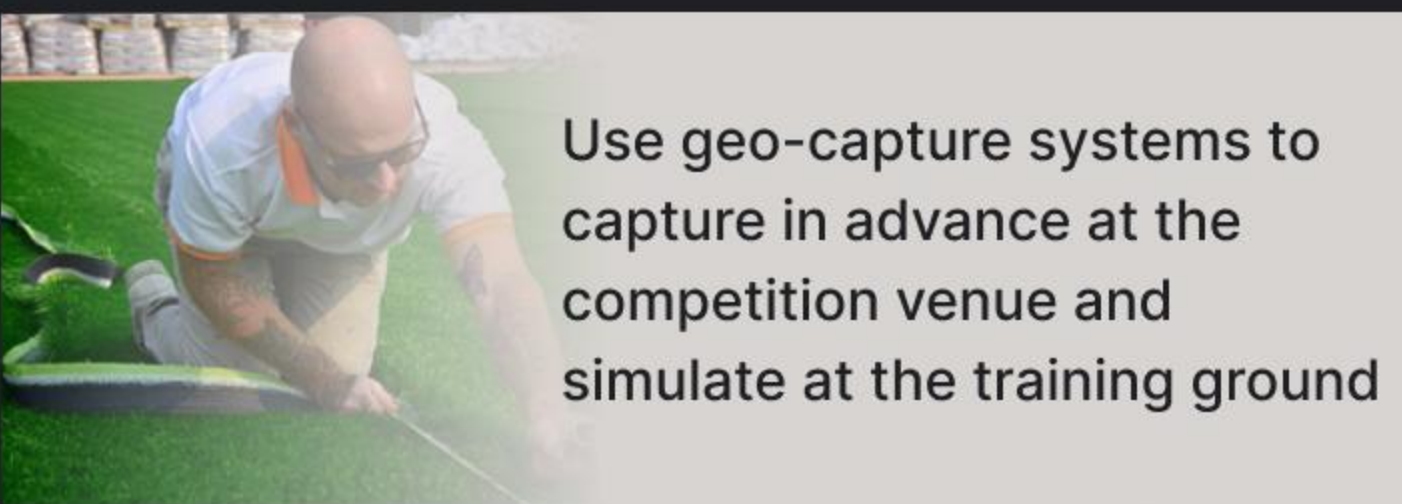
Fan Vibe



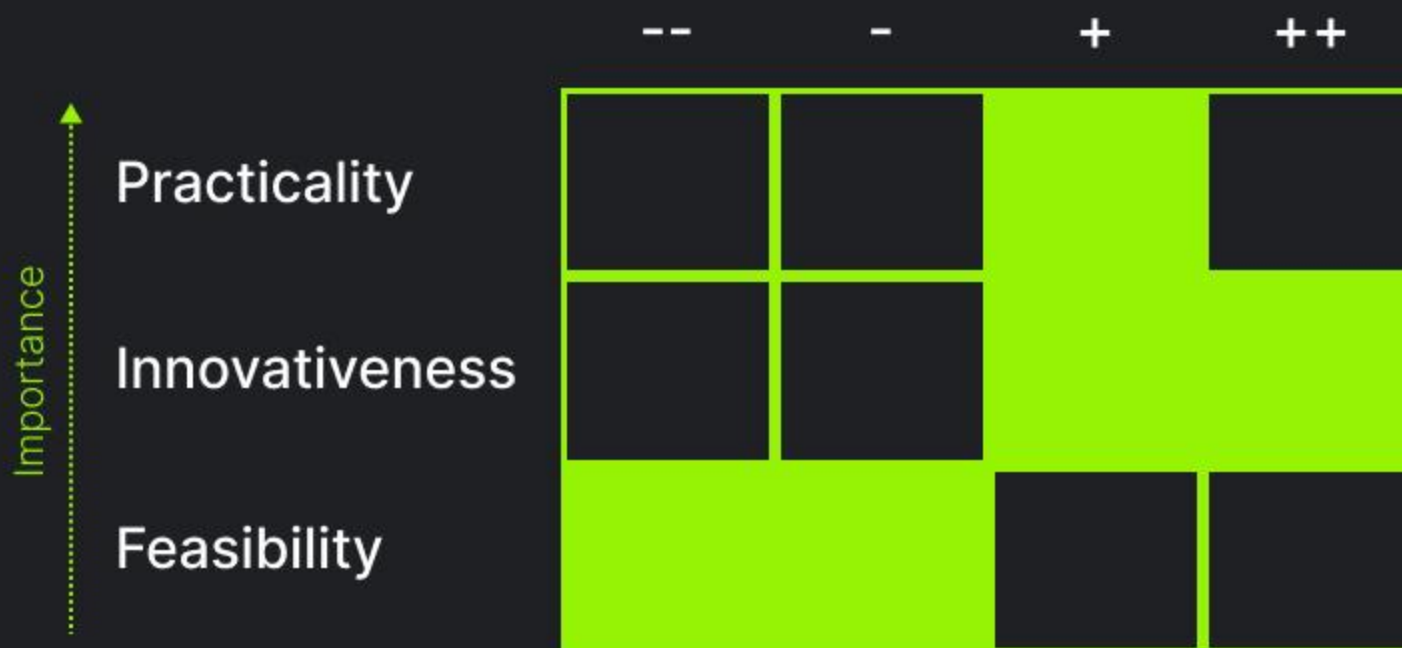
Simulate the sound of home and away fan atmospheres based on the position of the ball as it enters different zones



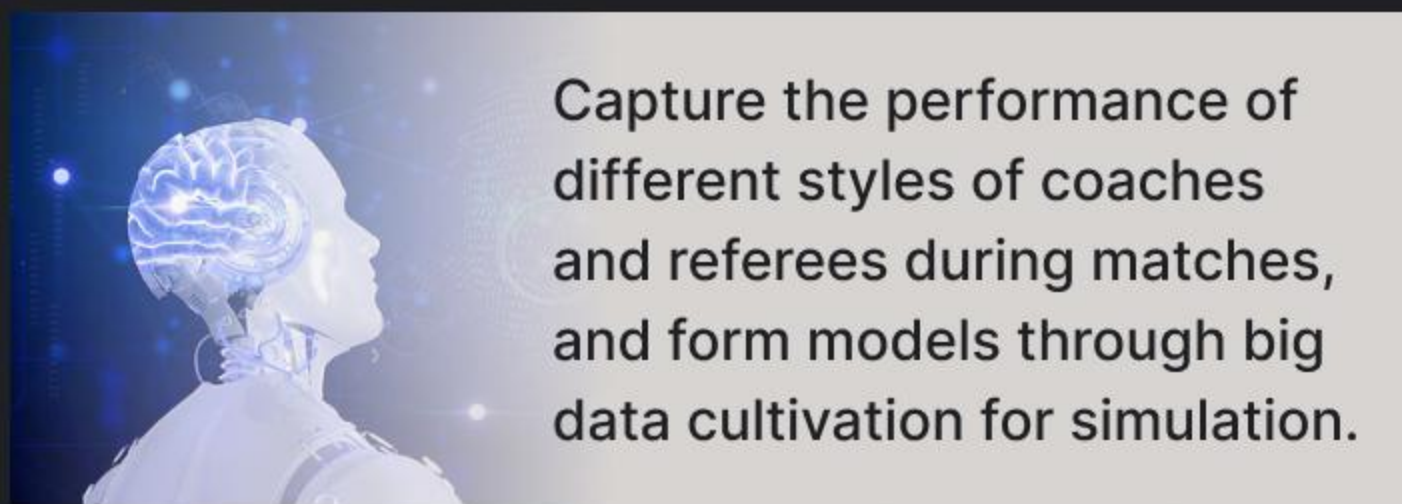
Football field facilities



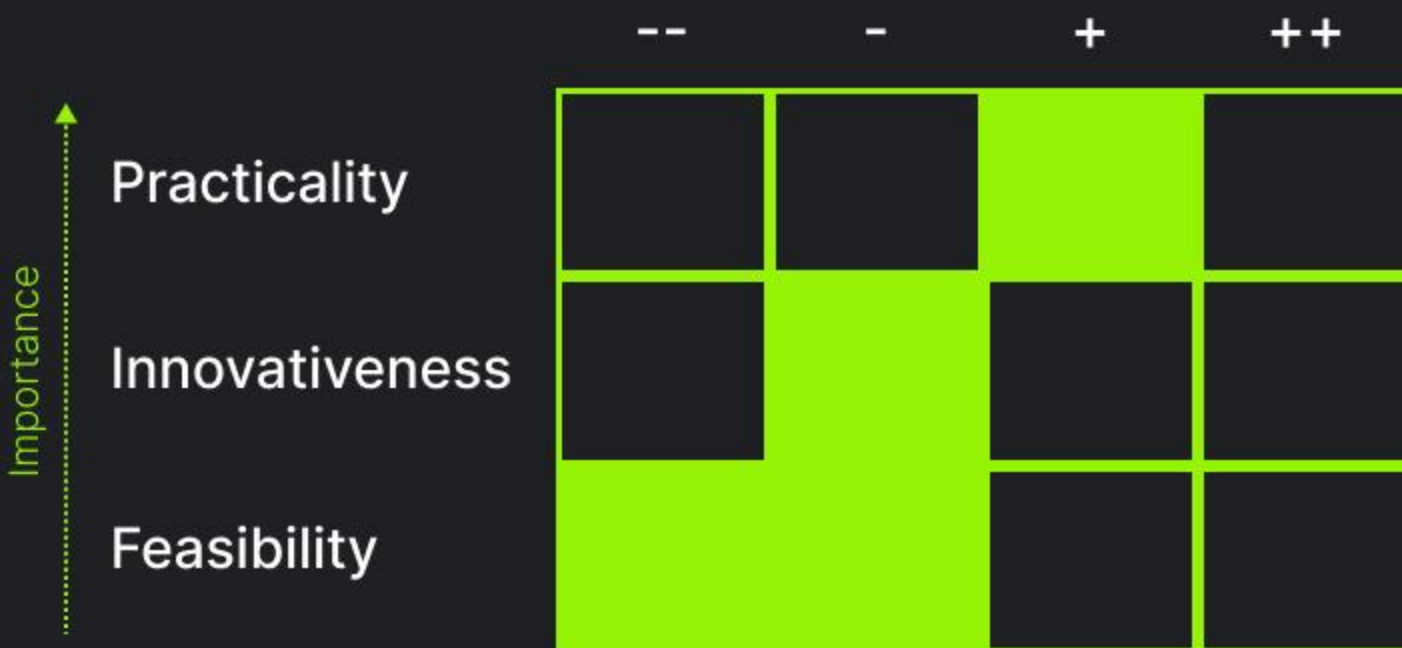
Use geo-capture systems to capture in advance at the competition venue and simulate at the training ground



Coaches & Referees



Capture the performance of different styles of coaches and referees during matches, and form models through big data cultivation for simulation.

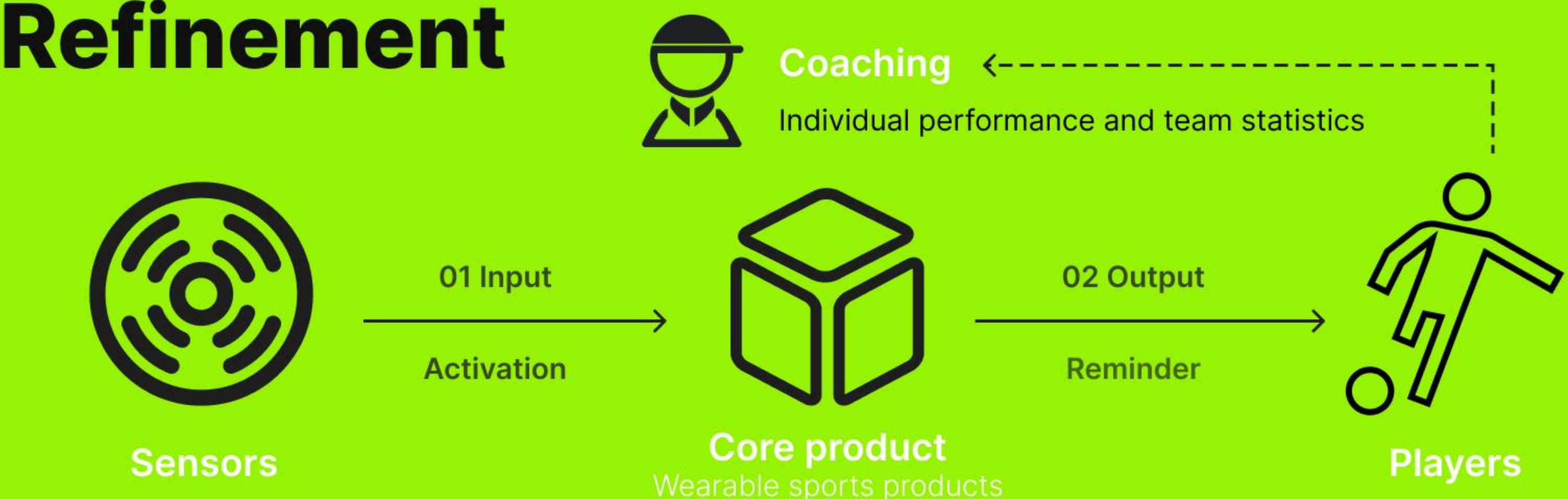


Practicality Can it be applied consistently without creating new problems

Innovativeness Whether the programme is innovative and refreshing

Feasibility Availability of current technology and complexity of operation

Refinement



01 Input-Activated Logic



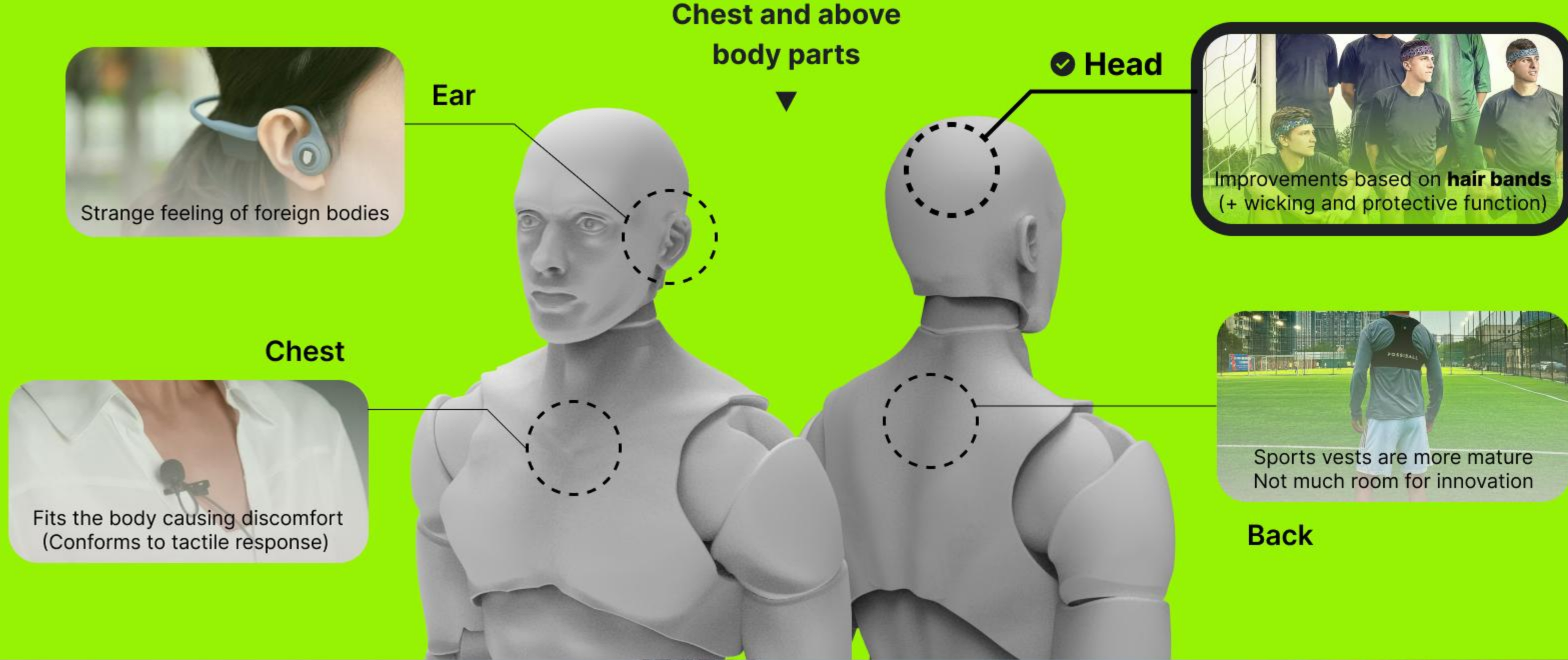
02 Outputs - transmission routes

Based on the motivation logic, the product needs a reminder that can be responded to in a **graded manner** and can **do so quickly and without creating a major obstacle** for the player.

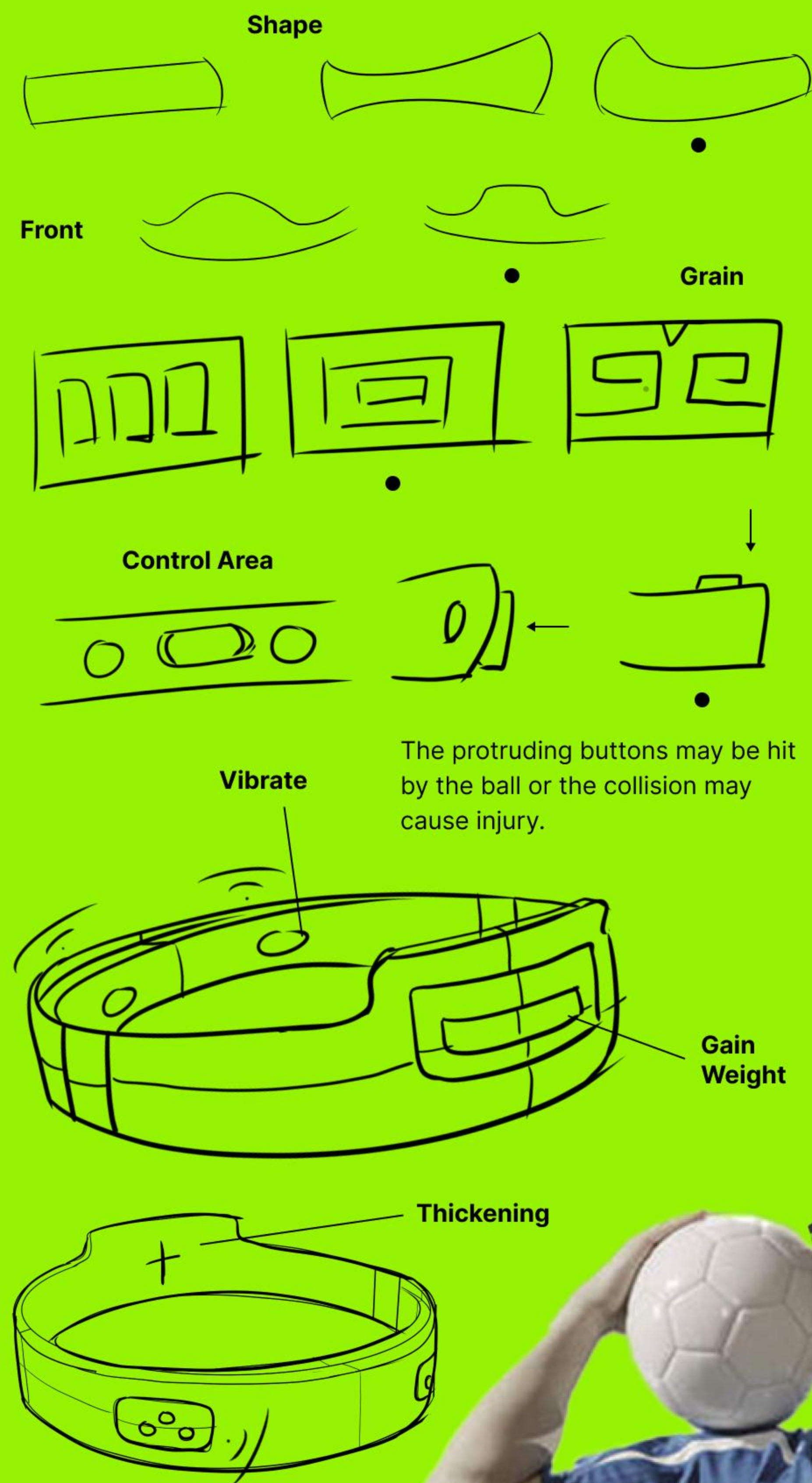
Modality of perception	Hearing	Taste	Olfactory	Vision	Touch	Intuition
Clearly graded	████████	████████	████████	████████	████████	████████
Rapid response	████████	████████	████████	████████	████████	████████
No interference	████████	████████	████████	████████	████████	████████



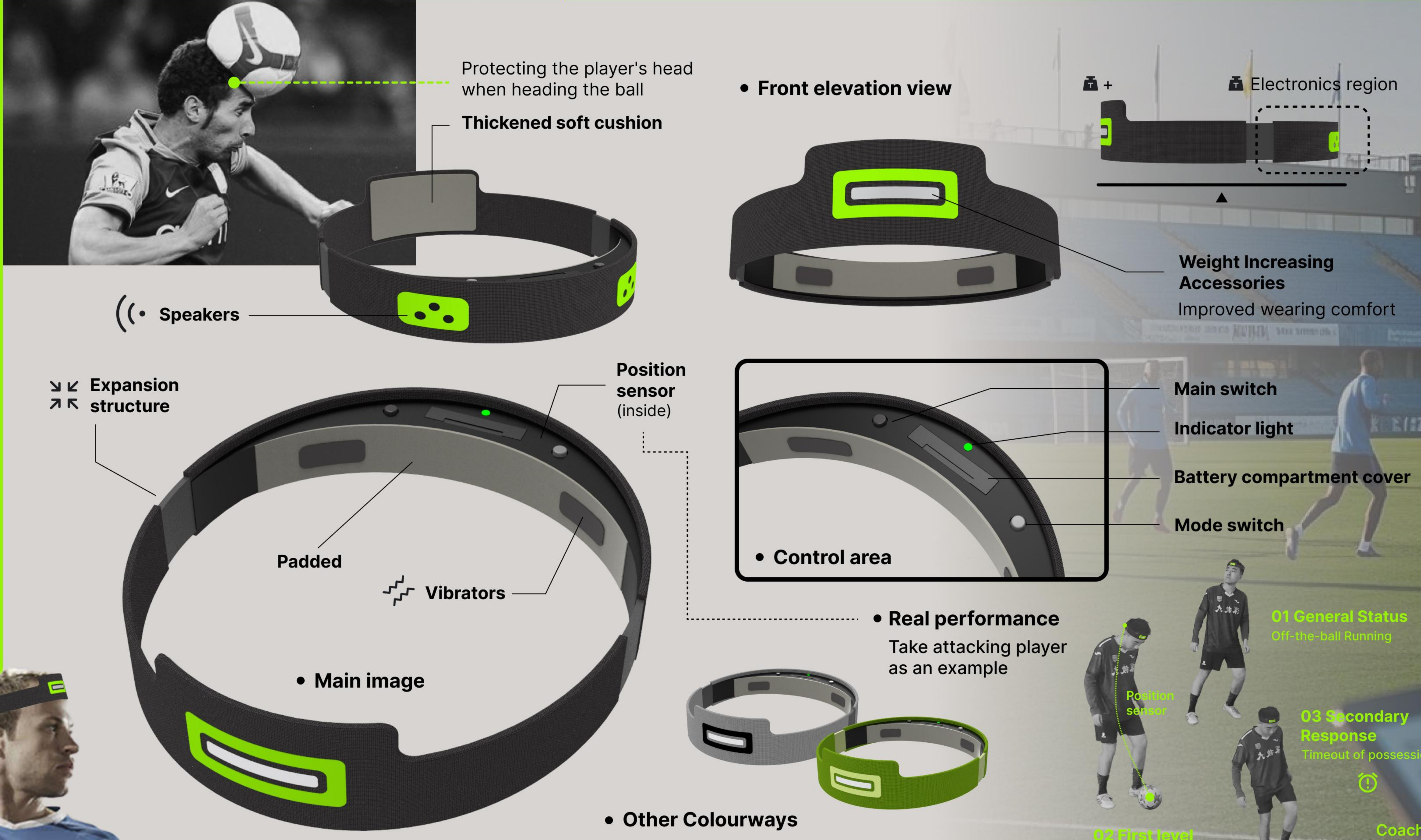
Adaptable parts Use current player equipment as an entry point to explore alternative or improved designs to reduce player discomfort and enhance simulation training



Design Showcase



Sketch



Product Effect

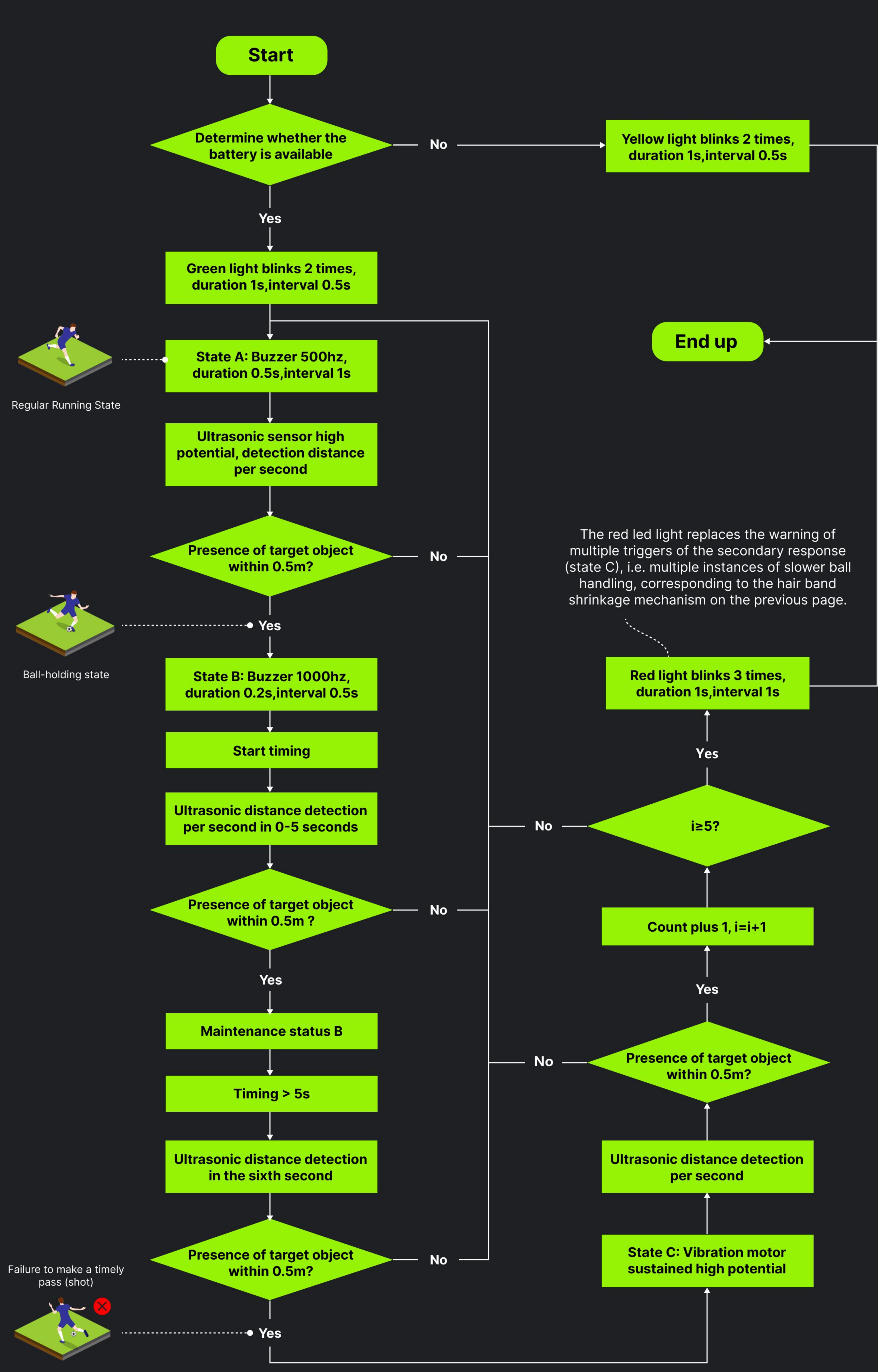


Operating state

Coach's Page

Coaches can analyse each player's ball handling speed and make adjustments, as well as set up rewards and penalties based on performance.

01 Project Flowchart



02 Write code

```
1 // Pin Definitions
2 const int trigPin = 3;
3 const int echoPin = 2;
4 // Passive buzzer connected to PWM pin
5 const int buzzerPin = 5;
6 const int vibrationPin = 7;
7 const int redLedPin = 10;
8 const int greenLedPin = 11;
9
10
11
```

Defining Pins

Outputs a PWM signal to adjust the volume by changing the duty cycle.

```
17 // Volume control
18 const int volumeLow = 100;
19 const int volumeHigh = 255;
```

```
9 // State Definition
10 enum State { A, B, C };
11 State currentState = A;
```

```
13 // Timer/Counter
14 unsigned long timer = 0;
15 int count = 0;
```

Define status/ timer/counter

```
20 // Pin Initialisation
21 void setup() {
22   pinMode(trigPin, OUTPUT);
23   pinMode(echoPin, INPUT);
24   pinMode(buzzerPin, OUTPUT);
25   pinMode(vibrationPin, OUTPUT);
26   pinMode(redLedPin, OUTPUT);
27   pinMode(greenLedPin, OUTPUT);
28
29
30 // initial state
31 digitalWrite(greenLedPin, HIGH);
32 delay(1000);
33 digitalWrite(greenLedPin, LOW);
34 delay(500);
35 digitalWrite(greenLedPin, HIGH);
36 delay(1000);
37 digitalWrite(greenLedPin, LOW);
38 currentState = A;
39
40 }
```

Initialisation pins

```
41 // Ultrasonic ranging
42 void loop() {
43   float distance = getDistance();
44
45   // change-of-state logic
46   switch (currentState) {
47     case A:
48       if (distance < 0.5) {
49         currentState = B;
50         timer = millis();
51       } else {
52         tone(buzzerPin, 500);
53         analogWrite(buzzerPin, volumeLow);
54         delay(500);
55         noTone(buzzerPin);
56         delay(1000);
57       }
58       break;
59
60     case B:
61       if (distance >= 0.5) {
62         currentState = A;
63       } else {
64         tone(buzzerPin, 1000);
65         analogWrite(buzzerPin, volumeHigh);
66         delay(200);
67         noTone(buzzerPin);
68         delay(500);
69       }
70
71 // Timer time verification
72 if (millis() - timer >= 5000) {
73   currentState = C;
74 }
75 break;
```

```
77 case C:
78   if (distance >= 0.5) {
79     count++;
80     if (count >= 5) {
81       digitalWrite(redLedPin, HIGH);
82       delay(3000);
83       digitalWrite(redLedPin, LOW);
84       shutdown(); // Shutdown Logic
85     } else {
86       currentState = A;
87     }
88   } else {
89     // Vibration motor work
90     digitalWrite(vibrationPin, HIGH);
91     delay(1000);
92     digitalWrite(vibrationPin, LOW);
93     delay(500);
94   }
95   break;
96 }
97 }
```

ABC status and switching judgement

```
99 // Get Ultrasonic Ranging
100 float getDistance() {
101   digitalWrite(trigPin, LOW);
102   delayMicroseconds(2);
103   digitalWrite(trigPin, HIGH);
104   delayMicroseconds(10);
105   digitalWrite(trigPin, LOW);
106
107   long duration = pulseIn(echoPin, HIGH);
108   float distance = duration * 0.034 / 2; // Calculated distance (cm)
109   return distance / 100.0; // Conversion to metres
110 }
```

```
112 // Shutdown Logic
113 void shutdown() {
114   digitalWrite(buzzerPin, LOW);
115   digitalWrite(vibrationPin, LOW);
116   digitalWrite(redLedPin, LOW);
117   digitalWrite(greenLedPin, LOW);
118
119 // Stop programme running
120 while (true) {
121 }
122 }
```

Get Ultrasonic Ranging

```
122 }
```

Shutdown Logic

Program stops running when all peripherals are switched off.

03 Circuit simulation

Problem

Simulation of the operation of the motherboard burned out, troubleshooting and analysis of the vibration motor power is larger

Solution

- 01 Vibration motor series diode control current. ✖
- 02 Vibration motor control using nmos transistors, powered using the 5V supply connector on the motherboard. ✔

Analysis

The vibration motor relies directly on the pin connection for power supply causing the motherboard to burn out.

04 Actual circuit connections

- Arduino Uno R3
- Dupont Line
- 5mm LED
- Breadboard
- 220Ω Resistors
- HC-SR04 Ultrasonic Sensor
- IRF540N Transistor
- Passive Buzzer
- Vibration motors

The buzzer and vibration motor wires are short and prone to poor contact, the problem is solved by replacing them with modular components.

Experience = Treasure

- Product design
- AR interface design
- Business design

Inspired by my interest in local history and a film called *The Internship*, I developed a project where seniors contribute to local chronicles, relieving depression and addressing financial needs. It is a pathway linking personal experiences to local history, enabling them to monetize their knowledge.

The final solution, inspired by *active aging* and technological trends, is an AR app that uses AI to convert oral histories into visuals, creating diverse formats for presenting local history.



Inspiration

'Lord Fan' statue

My hometown is known as 'Fan Gong's hometown', in honour of Fan Gong, the government built a statue as a city card.

1989 Establish

2003 Second relocation

1995 First relocation

2024 Current Location

People like to call the spot where the statue used to be **'Old Lord Fan'**, which often confuses me because it's a name for a place that isn't even marked on a map!

Local Chronicles's Problems

Social Dimension

Failure to form good social connections

Facility Dimension

Inadequate construction of working institutions & collections

Talent Dimension

Difficulty in data collection. Weak cultural identity

Experience never goes out of fashion

Robert DENIRO Anne HATHAWAY

INTERN

Plenty of time Cultural identity Personal experience Well-informed

Inspired by the film *'The Internship'*, I realised that **older people** can be a vital force in the construction of Local chronicles.

Just like the film's main theme.

'Experience never goes out of fashion'

Investigation

Oral history

In fact, this method of historical research, which collects the oral memories and personal experiences of individuals or groups of individuals, is of great value to historical research and is called **'oral history'**.

By observing my grandparents, their friends and my elderly neighbours, I found that an interesting phenomenon is that today's elderly people **spend a lot of time on their mobile phones every day**.

Relevant Data

As of June 2023, the size of China's Internet user group aged 60 and above had reached **140 million**. — From CNNIC

Nearly **10%** of elderly people go online for more than **6 hours** a day. — From Tencent and People's Daily

Phenomenon

Internet addiction

Nature

Mental disorder

As the **Tencent report** writes

"More than 100,000 older people across the country are showing extreme loneliness online."

Analysis

World

Globally, more than **14.1%** of persons over 60 years of age suffer from psychological disorders.

Depression is among the most common mental health problems, with a global population of depressed older people of up to **80,000,000**

Approximately equal to the entire population of Germany

China

27.2%

Globally, about a quarter of all suicides occur among people aged 60 and over.

23%

China's prevalence of depression in old age is 23% globally, and even reached 26% (2019).

Disease burden of depressive disorders in the elderly in China

Disability-adjusted life years **DALY**

In layman's terms, this indicator indicates the time stolen by the disease.

Academics often analyse the causes of psychological problems in older people from a number of perspectives

Inadequate social participation

Participation of older persons in the compilation of local histories to enhance social participation.

Life Review Theory (Butler 1963)

It helps the elderly address unresolved issues from earlier in life, alleviates anxiety, and effectively relieves depression. And it is widely recognized abroad.

Lack of care for the family

Many elderly people and their families lack awareness of mental health issues, so proactive care and support are essential.

Degradation of body functions

Negative ageing —1990s→ Active ageing

Influenced by Positive psychology and Postmodern thinking

Appeal to the elderly themselves not to be filled with despair because they are no longer young.

Calls on young people not to neglect older people just because they are no longer young.

Enlightenment

Designers need to address ageing with a positive mindset and can try to use bold and innovative approaches.

User research

Retirement is an important point for the shift in the level of social participation of older people. My grandparents did the same.

Retirement		
Information	Attempt	Result
My grandma Age: 74 Occupation: Textile factory worker	Cross Stitch	Give Up Exacerbation of visual impairment Leads to lumbar muscle strain
My grandpa Age: 73 Occupation: Beer company manager	Invest in the food industry	Give Up Consuming a lot of energy Disproportionate returns

User Interviews

Interviewer

- What do you get out of it?
- Any regrets?
- Will you adapt to new technology?
- Other suggestions?

Interviewee

- Economic benefits
- Government Recognition
- Lost friends
- Old House
- I will learn and keep the tradition.
- Don't get tired of using it.

Summary

- People tend to look for opportunities in areas they are familiar with after retirement.
- Currently there is a lack of a home-based and healthy model of social participation for retirees.
- Ideally, design a business cycle that can generate some revenue for older people.
- Older adults adapt to technology but maintain long-held habits.

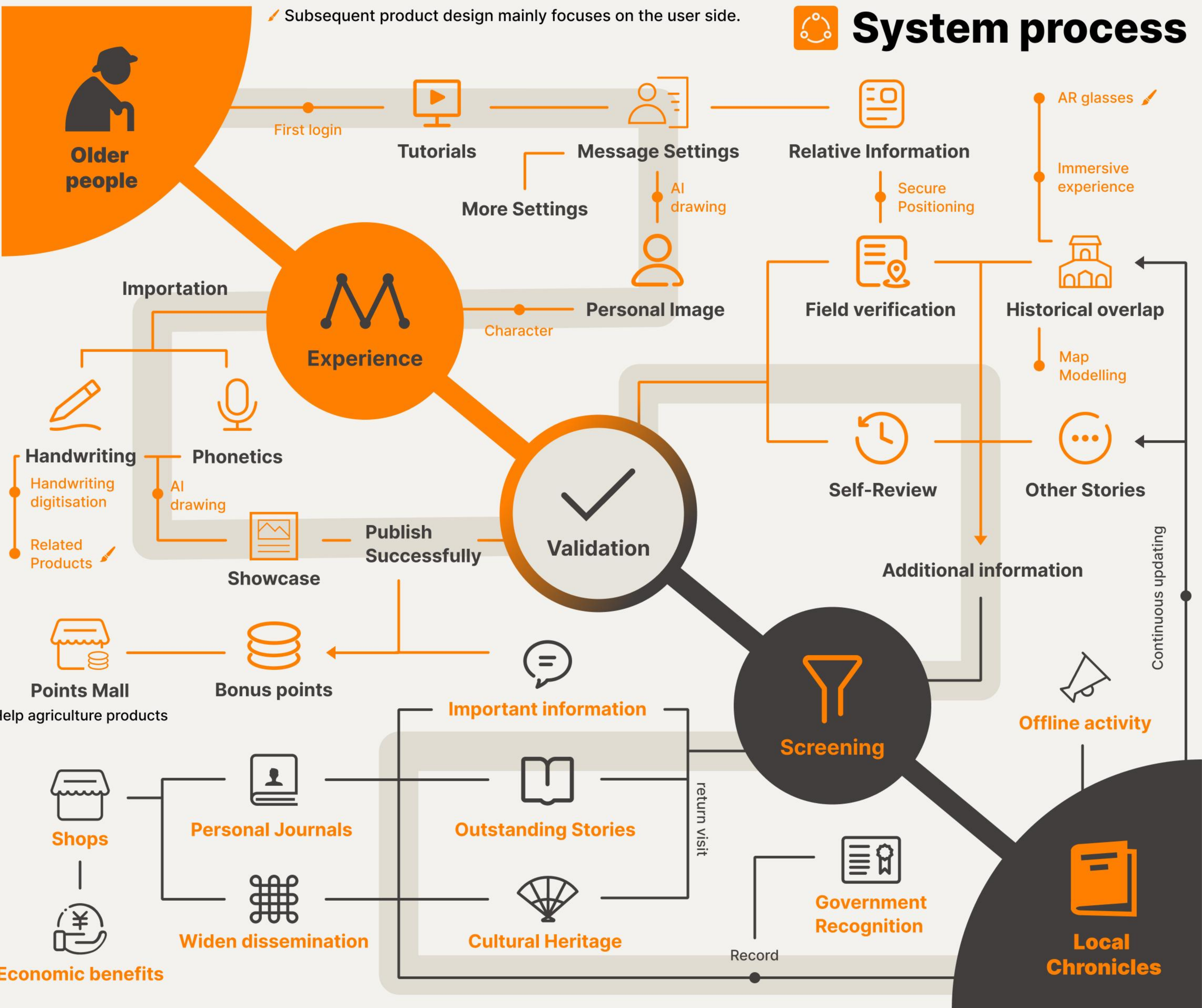
Definition

Products that connect local chronicles with the spiritual enrichment of older adults through the Life reviews theory and the Oral history while bringing additional value to the user.



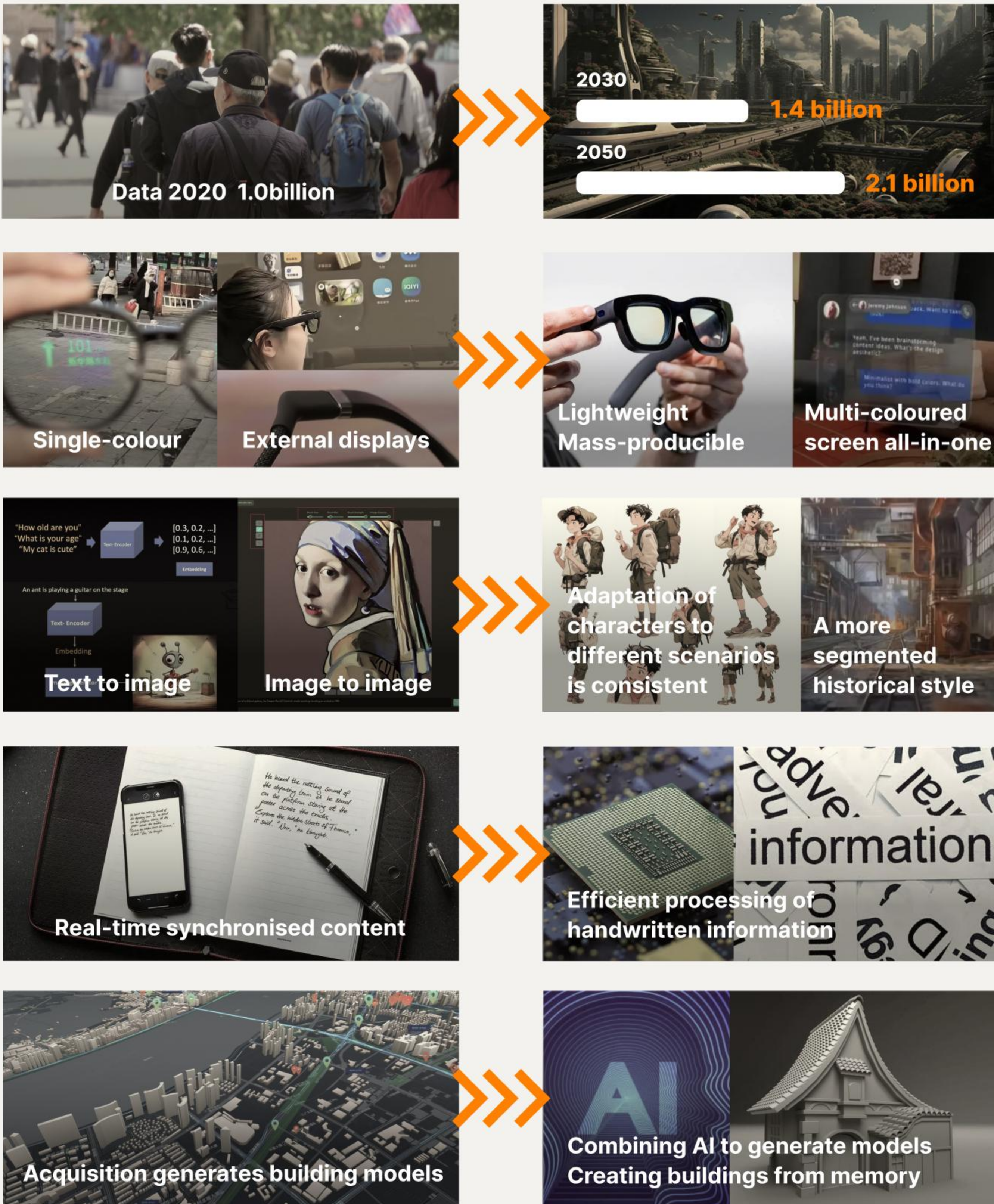
Business Analysis

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
Equipment manufacturer Government organisations Map Development Companies Publishers and distributors Agricultural dealers Historical researchers	Collecting experiences and verifying them Extract core into revenue Key Resources Creative forms Wide range of information Designed for the Elderly	By collecting the experiences of older people, it is possible to enrich history and bring many benefits to older people, while at the same time turning information into revenue and also creating a sharing platform for older people.	Make a return visit Organise offline events Personalised coaching Channels Social networking platform Sensitisation of children Government Co-operation	Recently Retired Senior Citizens Users who are not comfortable going out Children of the elderly History buffs
Cost Structure			Revenue Streams	
Equipment R&D User incentives	Production Costs Advertisement Costs	Labour costs Legal protection	Advertising revenue Sales commission	Sale of information Government Subsidies



Feasibility analysis

As the current technology is not perfectly adapted, the technology used in this product is based on the current development trend of short-term future applications, with a certain degree of foresight and feasibility.



Products

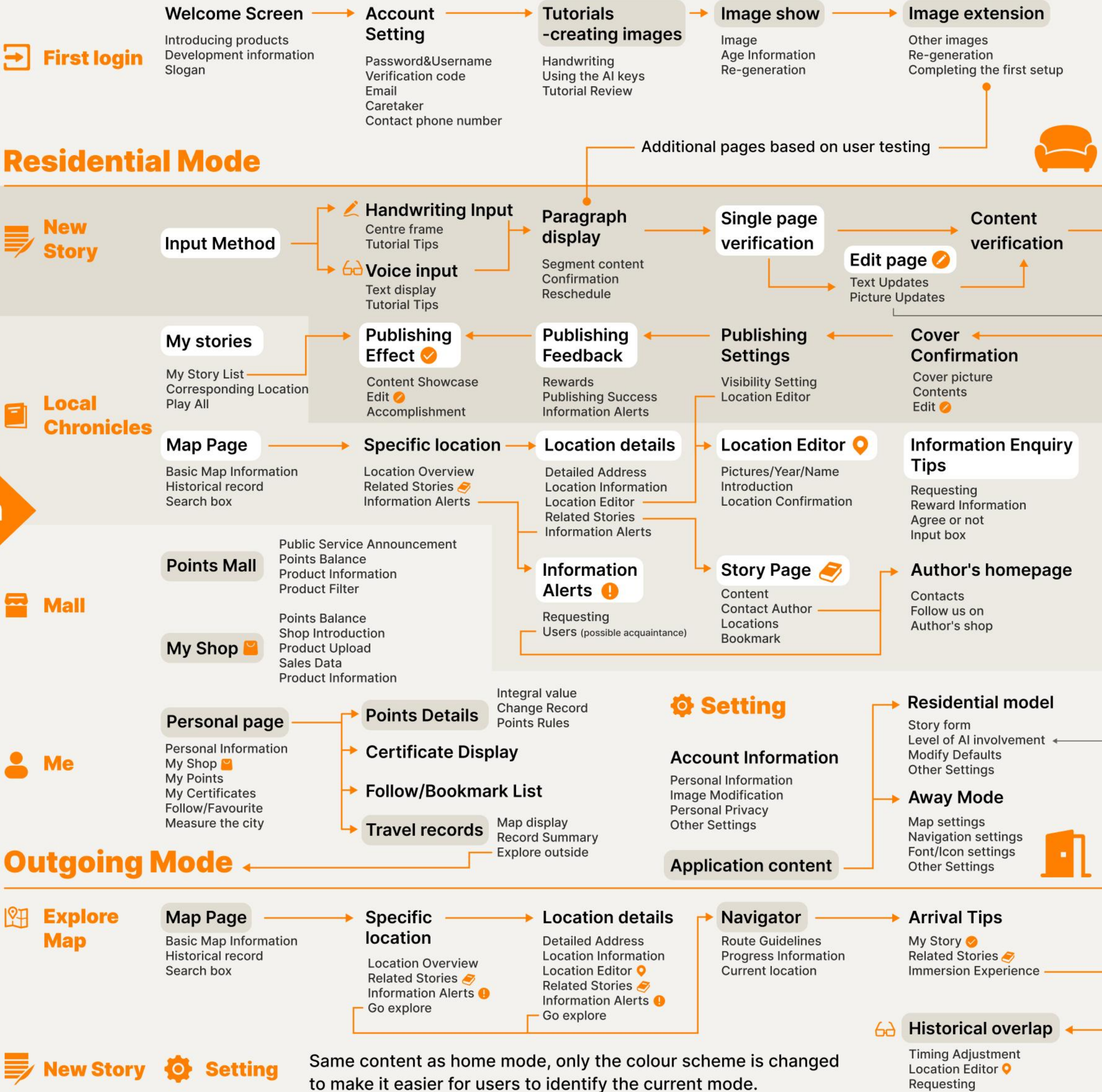
Hardware Products List

- 01 Pen (Handle)
- 02 Electronic handwriting book
- 03 AR glasses
- 04 Storage box



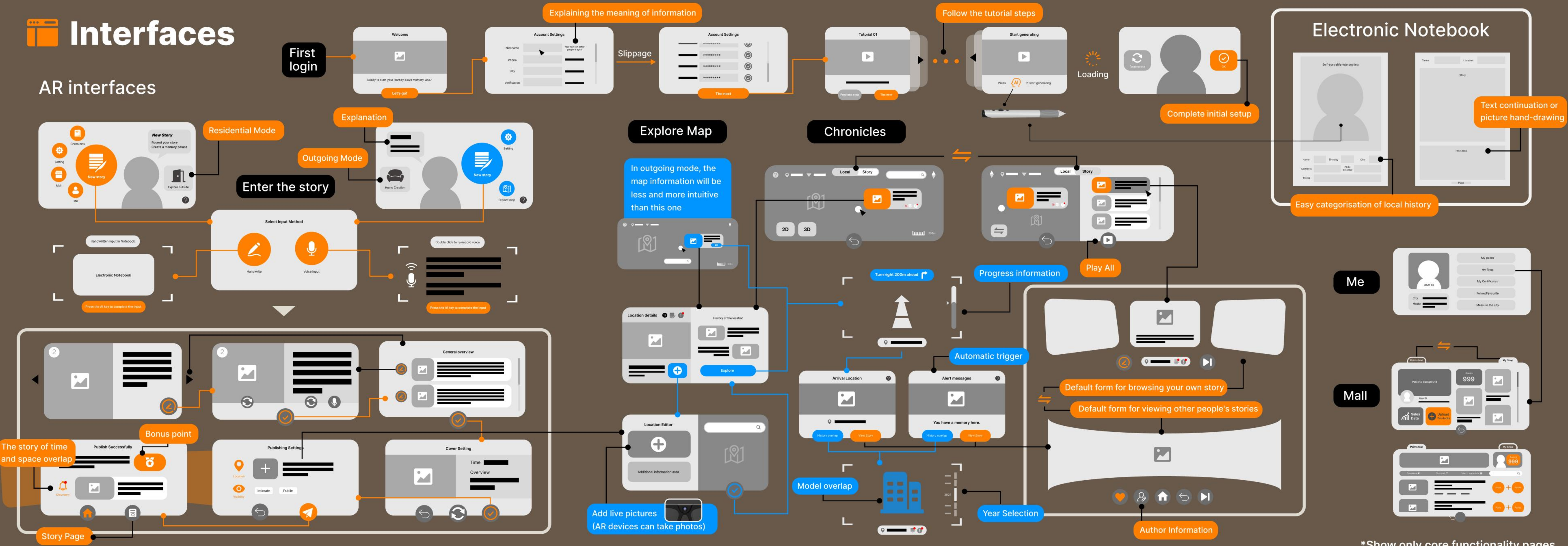
UI Framework

*Note: Since some of the links between the function blocks will cause overlapping, which will lead to logical confusion, the same icon will be used as a link in this case.



Interfaces

AR interfaces



Develop norms

Principle

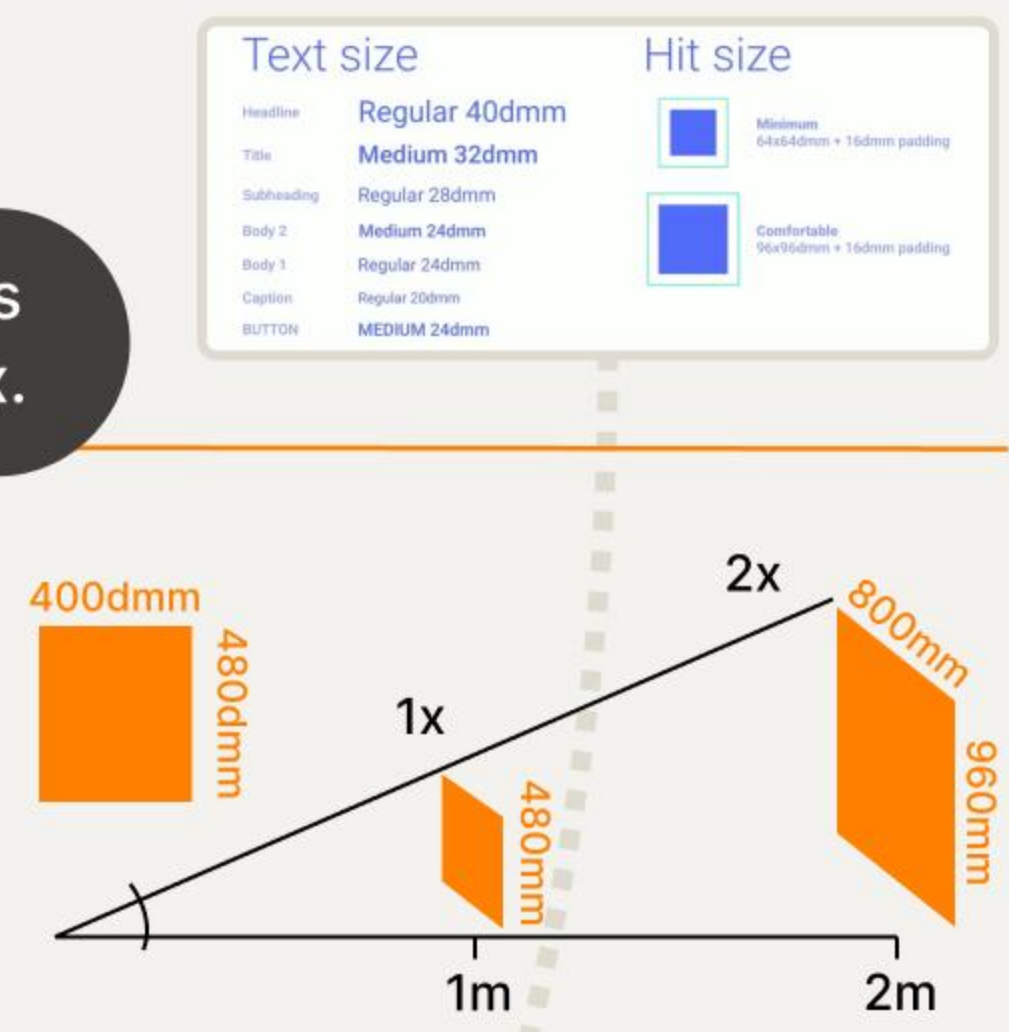
dmm

Distance-Independent Millimeter

The Meta team developed the 'dmm' concept, which forms the underlying specification for the UI design of XR products.

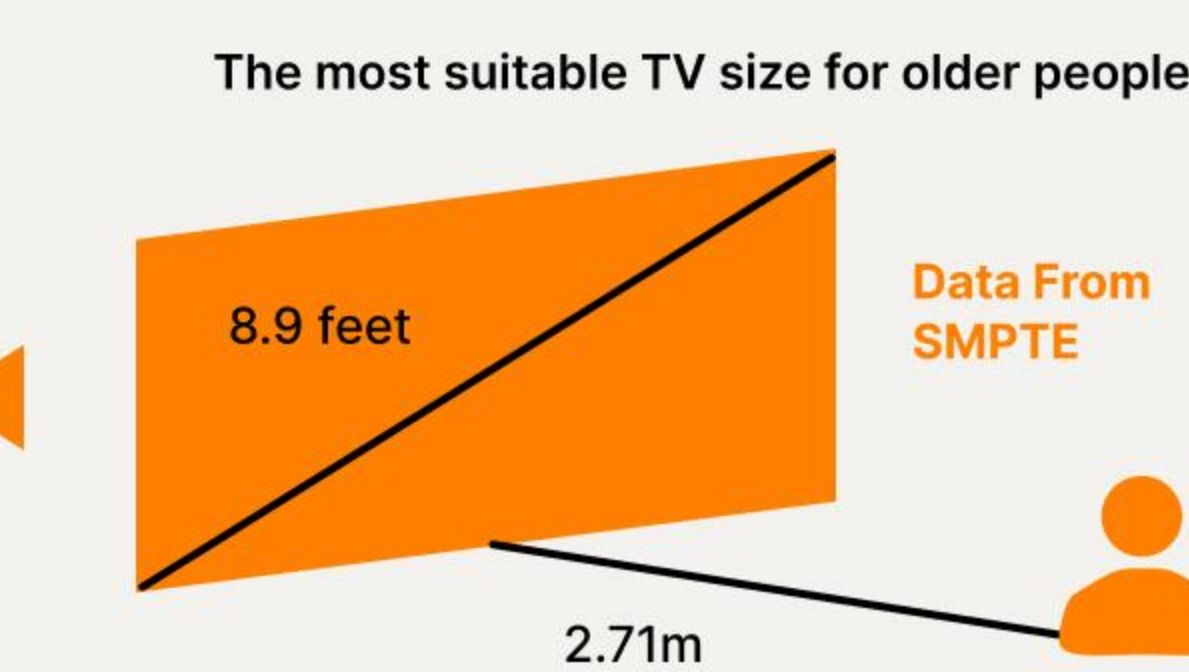


Both of the above cases correspond to 1dmm, which allows for consistency of content across different distances and screen sizes.



The same principle applies when expanding to the screen.

- ### Workflow
- Test the most suitable screen distance
 - Extrapolation of interface size by combining the dmm principle and known TV data
 - Test the smallest acceptable text size and icon size
- *Combined with Meta recommended sizes



This translates to a television set with a length and width of approximately 236cm and 133cm.



The size of the interface can be tailored to the TV viewing habits of older people.

Testing



Results

Consistent with the Oculus team's test results!

Distance 1.0m

236cm = ?

2.71m = 1.0m

Interface size 87×49cm

Text size 25dmm

Icon size 50×50dmm

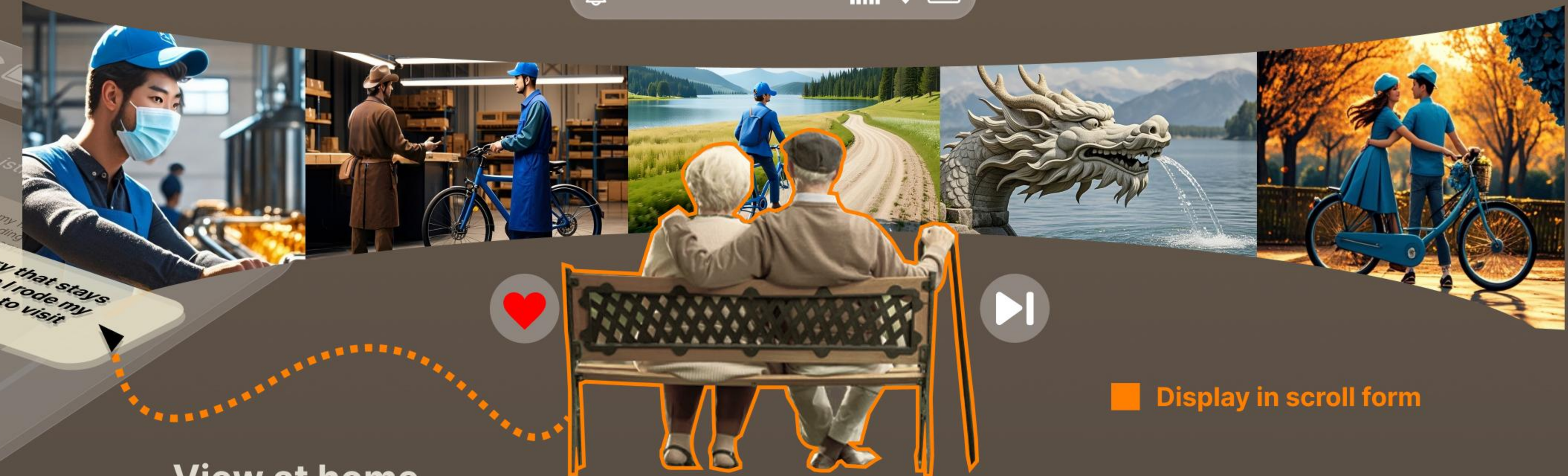
*The result is the minimum size of the design, which is used as the bottom line for the actual design.

Showcase

Homepage Residential Mode



Homepage Outgoing Mode



View at home

Display in scroll form

In order to prevent vertigo in the elderly, the focus during this type of presentation is centred on the back of the head, in line with the principles of retinal imaging.

Initial setup

Enter the story

Field exploration

Economic values

Historic value

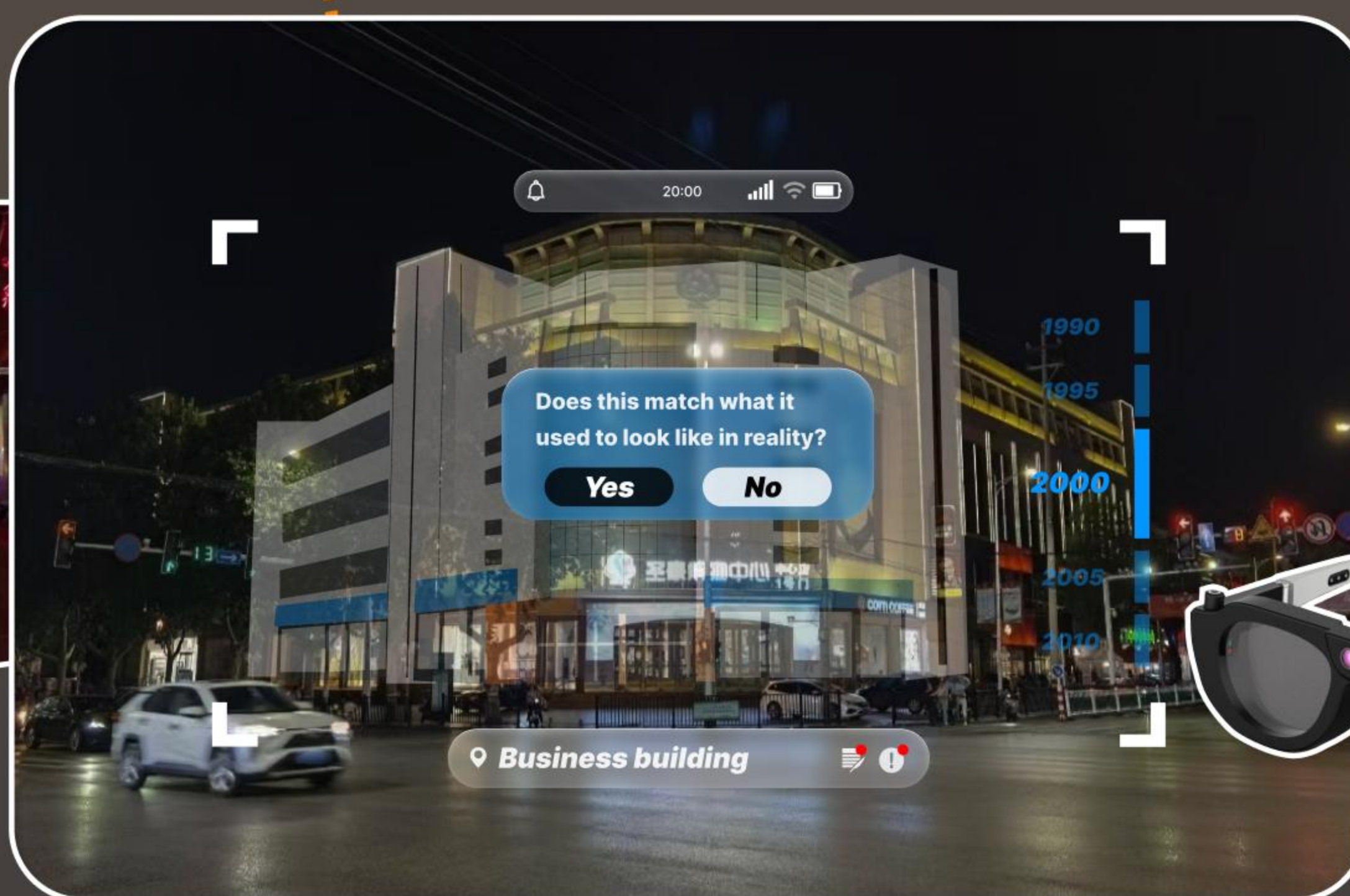


Auxiliary operations

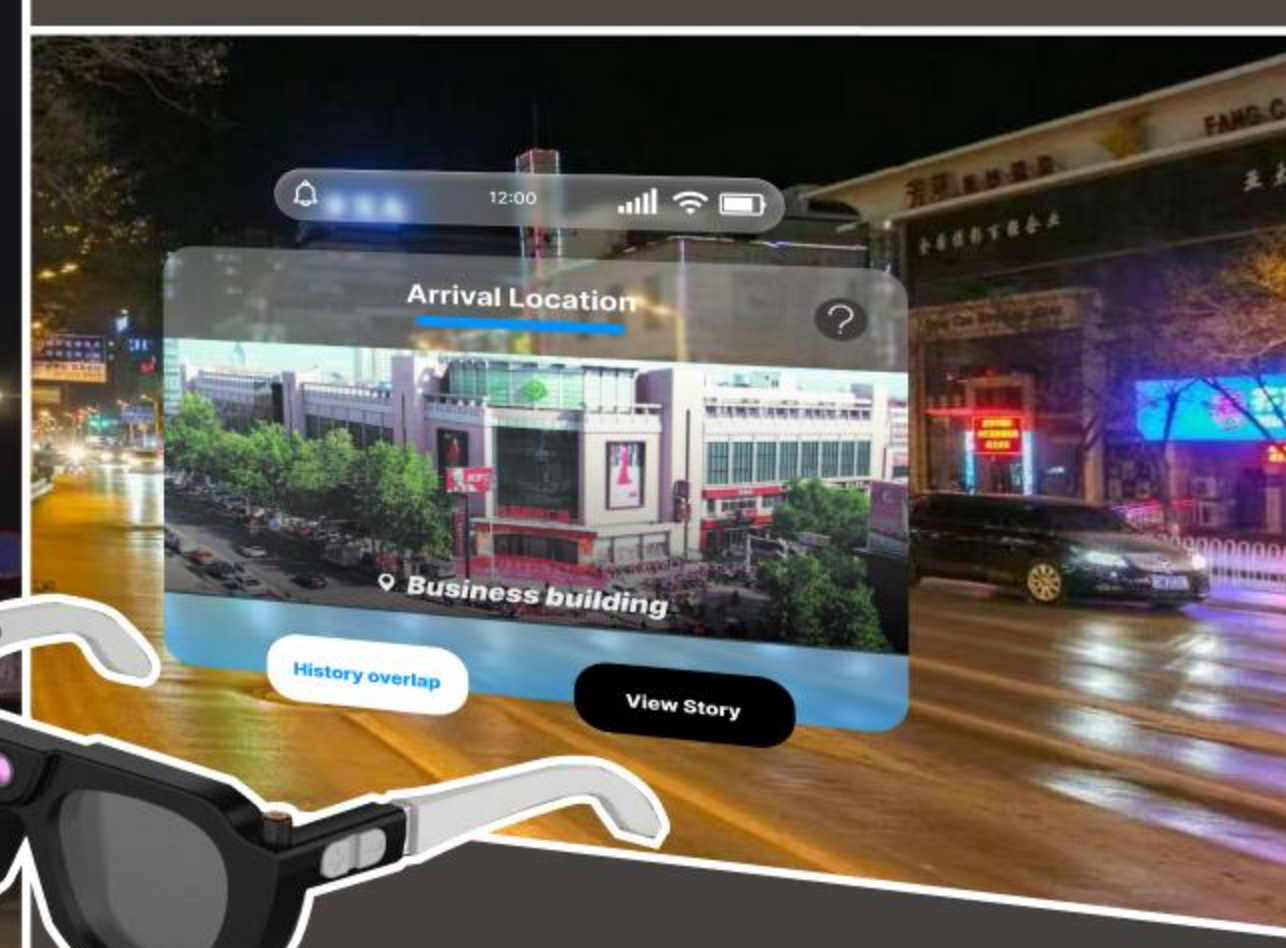
In addition to the selected buttons being spatially drawn closer together, a prompt message floats on the buttons of the handle.

Edit page

Navigator



Arrival Tips



Historical overlap



Online Mall

According to the point system, people can redeem the aid products. At the same time, for some intangible cultural heritage can be sold to promote culture. For ordinary users, they can sell their stories in the form of personal journals.

Local Chronicles

When the information is enriched in large quantities, the local history can be presented in a variety of forms.



The road to rebuilding confidence

- Service Design
- Product Design
- User Experience

An experience in high school led me to give up my dream of becoming a professional athlete, and now I want to do something for my younger self. Through literature research, I identified my condition as an episode of Functional Movement Disorders (FMDs) and learned that recovery requires comprehensive treatment to rebuild confidence.

Using a systematic service design perspective, I developed a home recovery program for teenage FMDs patients. The core solution features progressive treatment devices, supported by an integrated app for device control, community engagement, and psychological support.



Inspiration

This is something I experienced first hand, in a way this event changed my life, I will always remember it was late autumn 2017...

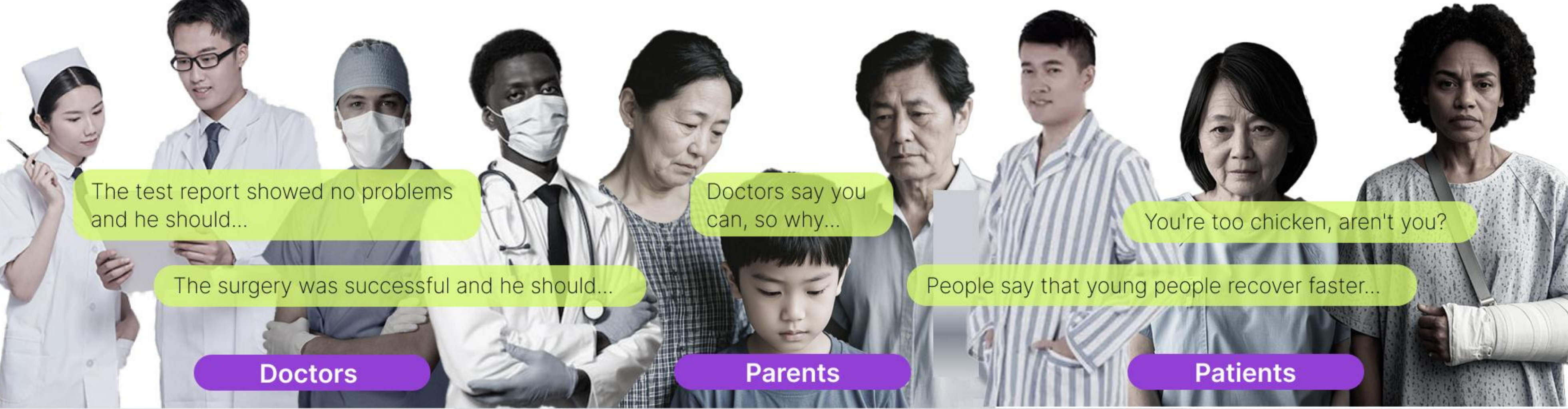


It was the sports day and I was participating in a 4*100m relay when I suddenly felt a sharp pain in my lower back and then fell down on the track

I was quickly taken by ambulance to the hospital, fortunately it was only a small bone pulled from a muscle and required a minor operation.

The day after the operation, I suddenly realised that I couldn't move my leg in any way, it was like I had lost all control of it!

Strangely, medical tests proved that I could lift my legs, so people around me were puzzled.



Eventually I was able to lift my leg, but I ...

- Gave up the dream of playing professionally
- Becoming less confident in life
- Become silent in character

“ Later in life I've been trying to do something about myself that day, and after some research I found out that it's actually a mental disorder called **FMDs (Functional Movement Disorders)**. ”



Literature Research

In conjunction with personal experience, this project focuses on **adolescent patients**.



FMDs (Functional Movement Disorders)

Simply put, FMDs are psychogenic movement disorders with **psychological factors** as the central trigger, but physical function tests are normal



My situation

In cases like mine, which start with a sports injury and are actually caused by a combination of psychological factors (fear, low self-esteem, worry, agitation...) athletic disorders are also in its scope.

Clinical Data

'FMDs are not rare among adolescents, with a gradually rising prevalence.'

Analysis of FMDs patient data from 1968 - 2019 showed **6.8%** were adolescents. Other studies reported FMDs patients could make up around **10 - 20%** of movement disorder cases in paediatric neurology outpatient clinics.

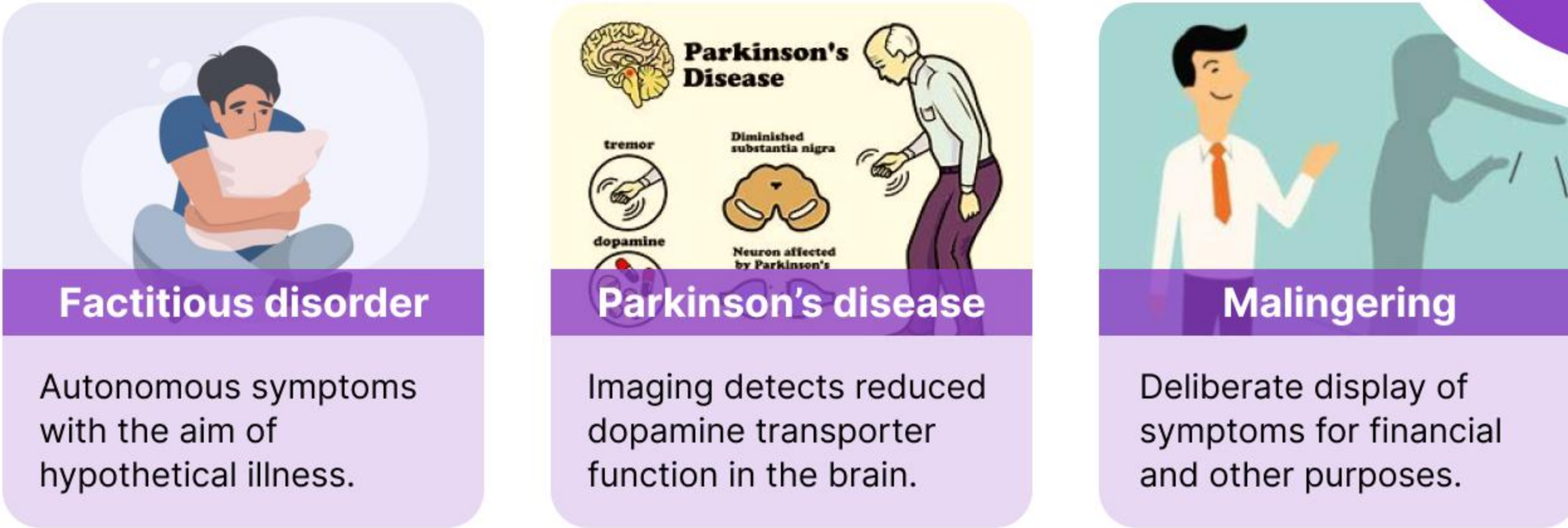


Most common clinical manifestations of FMDs in childhood and adolescence

I suffer from lower limb muscle weakness as a result of this clinical presentation, and this project will focus on this presentation as well.

*It can be detected with the Hoover sign, a classic sign for diagnosing functional weakness of the lower limbs

Clinical diagnosis requires caution to differentiate from similar diseases.



Distinguishing features - Sudden onset and abrupt cessation

'While most movement disorder disorders progress slowly, symptoms in FMDs occur suddenly, can be triggered by events such as trauma or illness, and can dramatically remit, or fluctuate in symptoms during the course of the disease.'

Disease characteristics

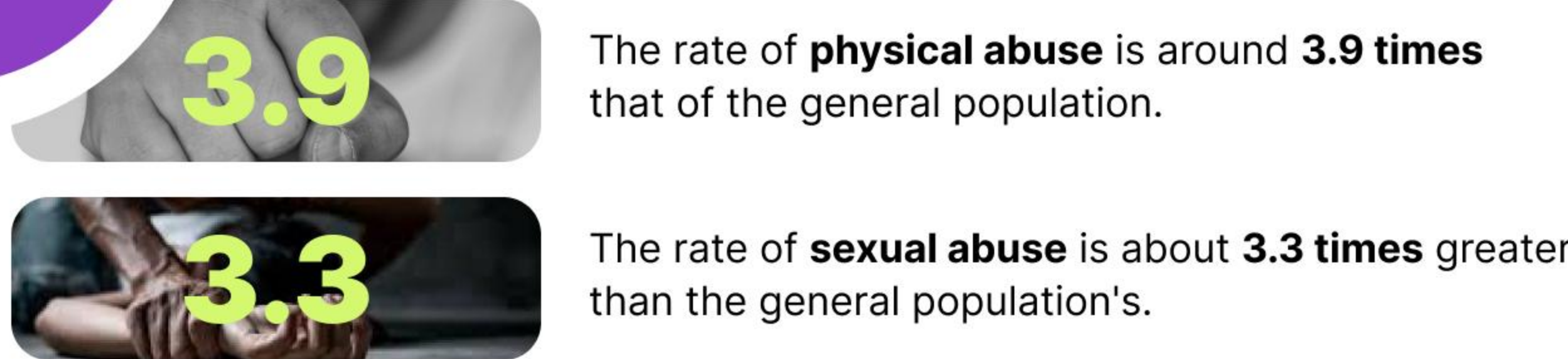
There are numerous triggers associated with adolescent patients with FMDs



The project focuses on such triggers (Personal Experience)

Sports injury type of causation + Combined psychological factors → FMDs

Research data reveals patients usually have a history of childhood trauma.



Boy 16 years old Student Quiet and a bit withdrawn Medium grades

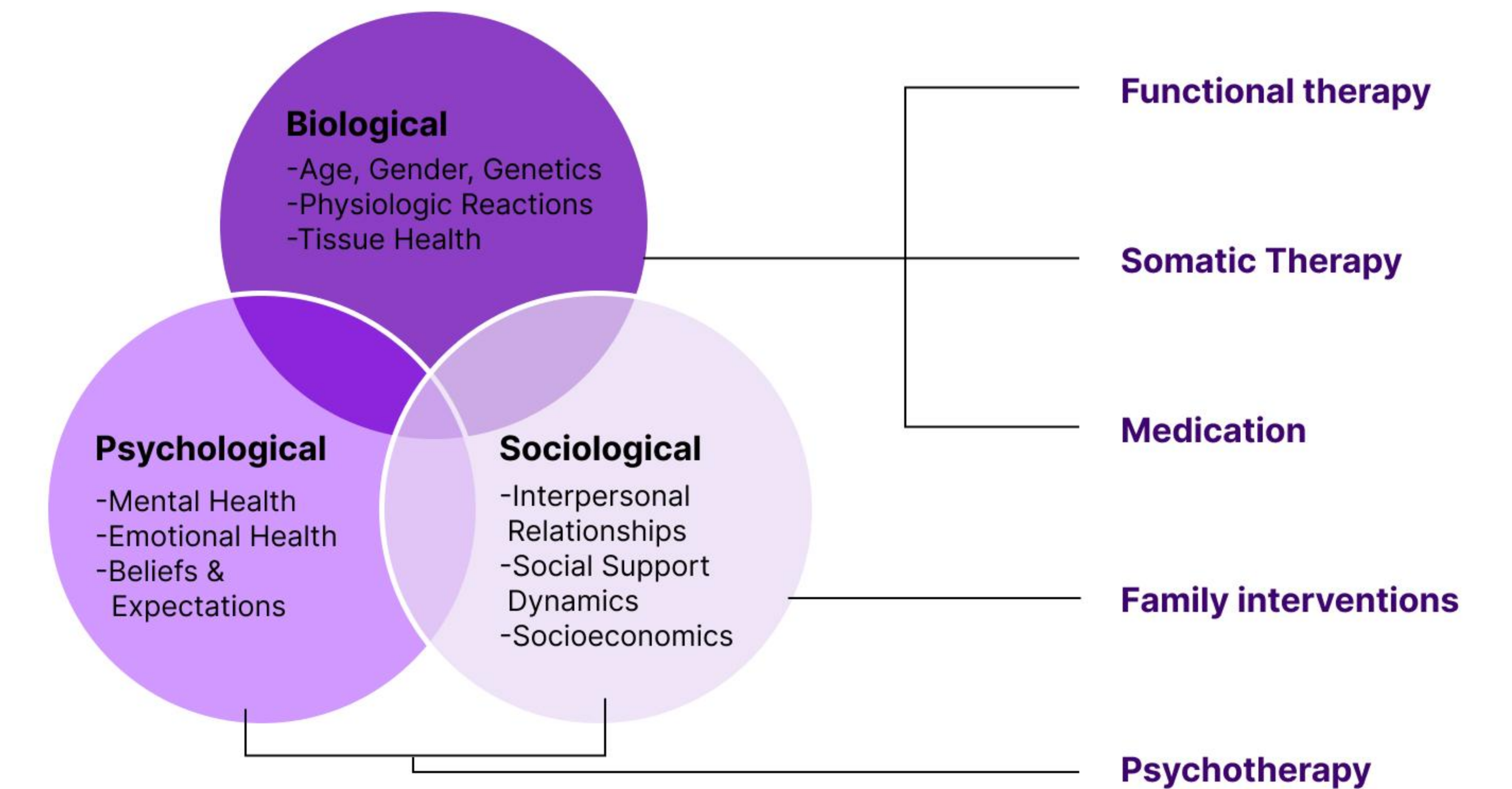
Patient (User) studies

Theory & treatment

The current consensus is to adopt a multidisciplinary model for treating this disease as its pathogenesis adheres to the BSP model.

BPS model (biological - psychological - social model) → Systematic service design

The traditional biomedical view deems biological indicators as the final determinants of disease, while the BPS model embraces the patient, disease, and environment. Physicians thus have to take into account the roles of social and psychological factors in addition to biological ones.



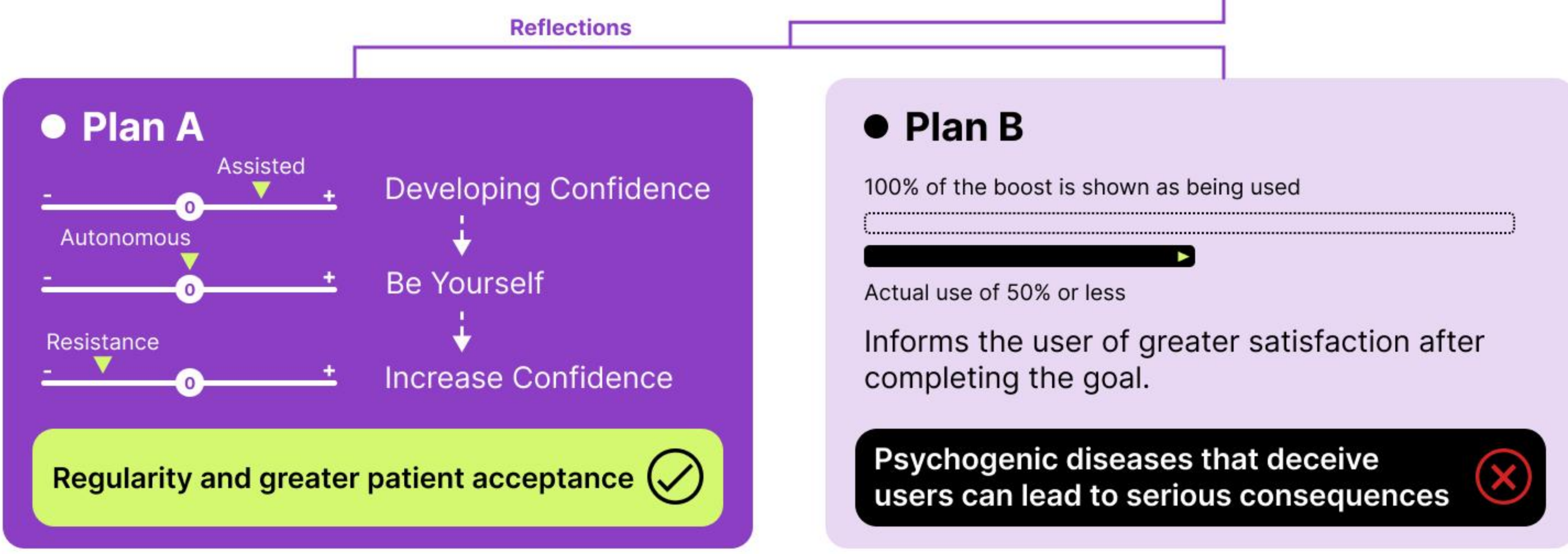
Programme planning

'Many studies have highlighted the challenges of developing standardised treatments for functional movement disorders and the need for individualised treatments tailored to the patient's clinical symptoms.'

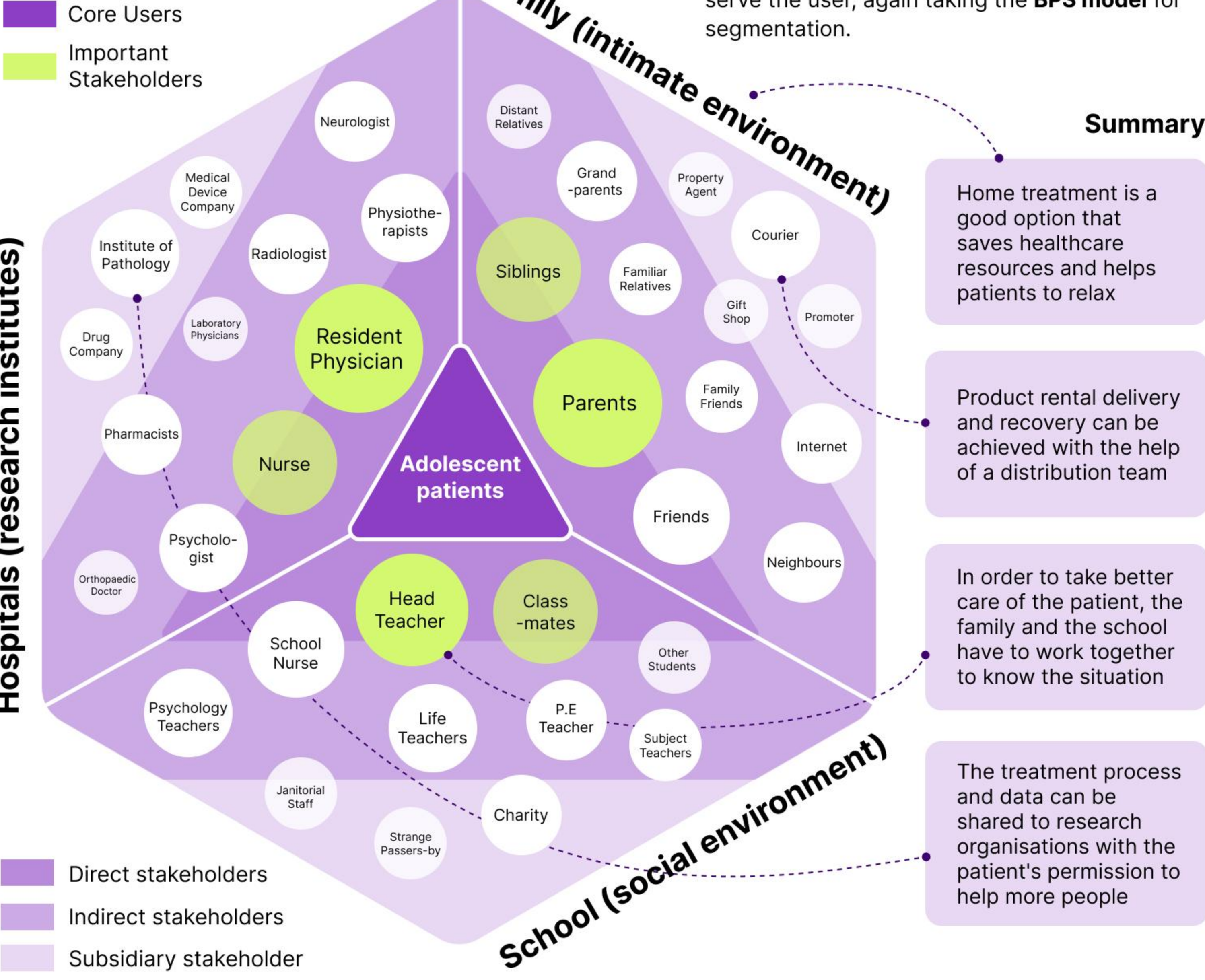
Customised and exclusive plans, personalised and timely assistive devices that allow both doctor and patient to share their statuses

'Physiotherapy and biofeedback focusing on progressive exercise training is helpful for most patients'

Therapeutic devices with progressive logic that can be micro-controlled



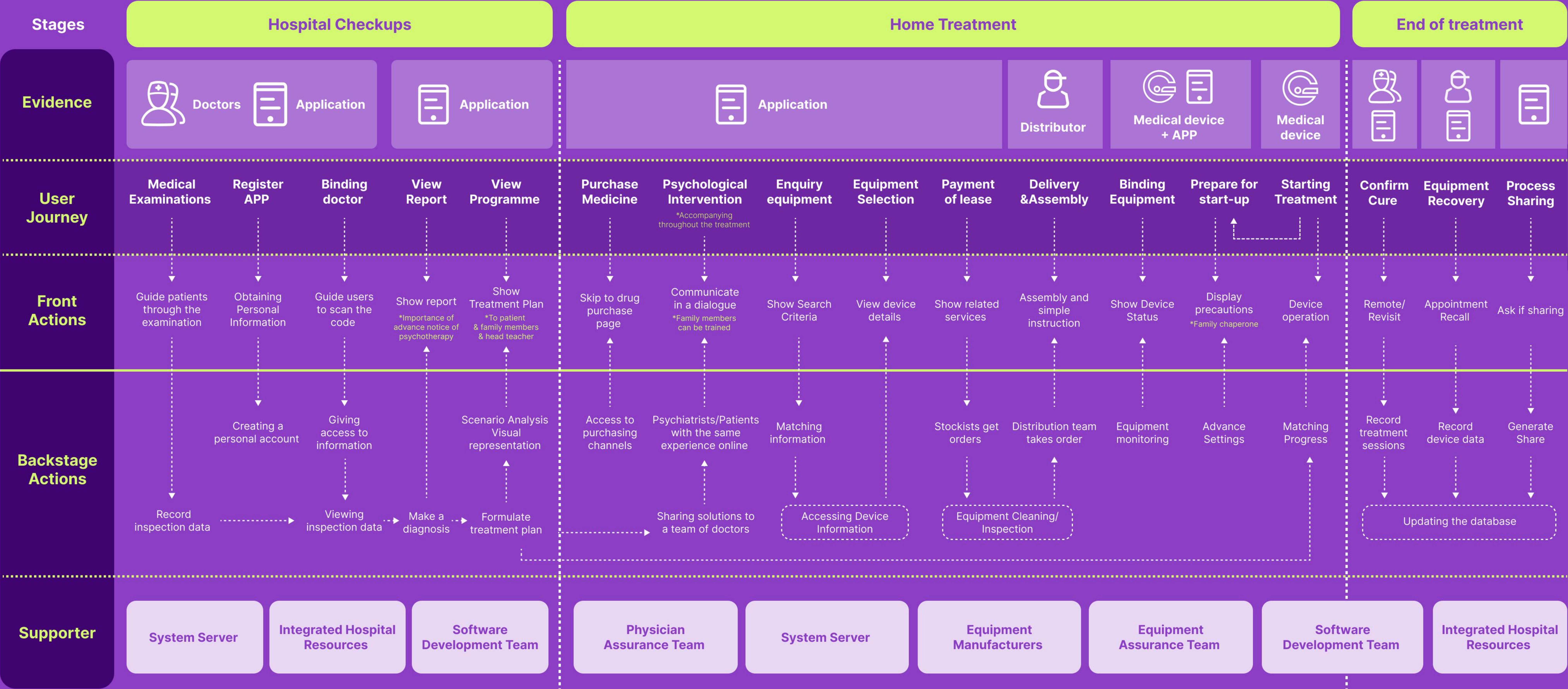
Stakeholders



Definition

Overview	Products	Scenario	Hospitals
	Micro-control hand and foot assistive mobility devices		Pre-programming and preparation
	Application	Reason	Home
	01 Product Rental and Ancillary Uses		Medical Products + App Assisted Therapy
	02 Psychological interventions		
	03 Therapeutic communication between doctor and patient		
User	Patients with FMDs in which organic damage is the causative factor	Time	Treatment (post-operative) recovery phase, once a day

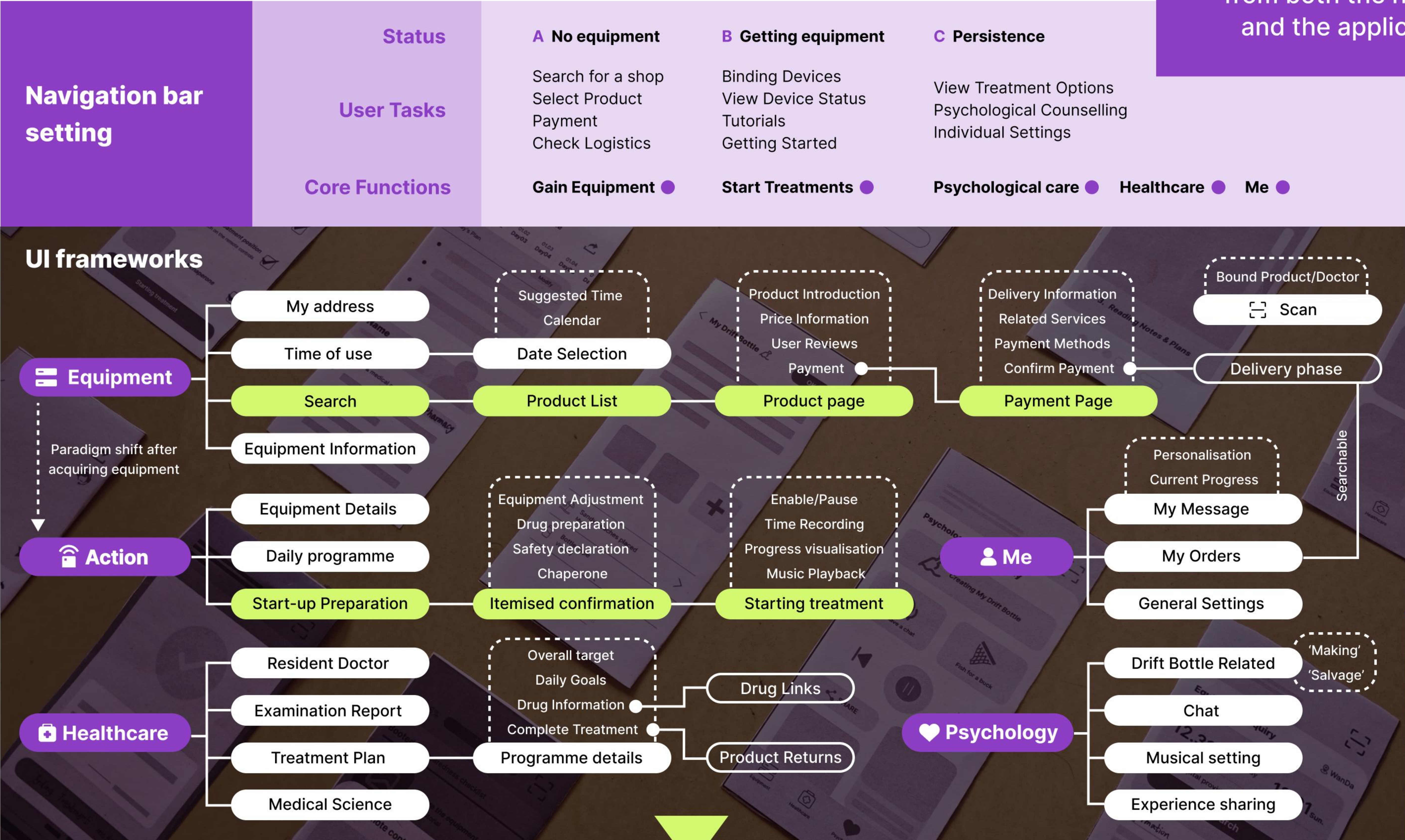
Service Blueprint



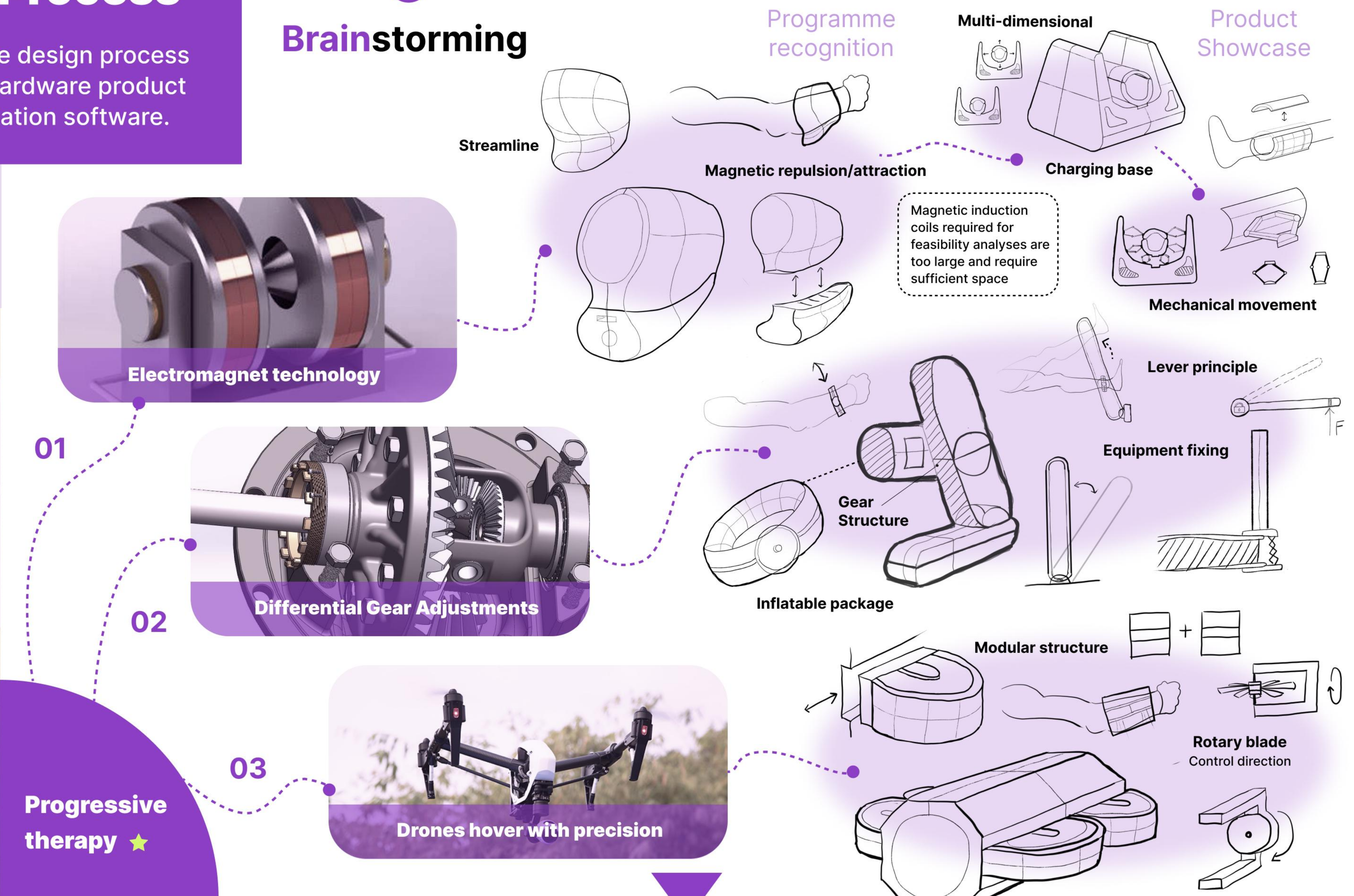
Design Process

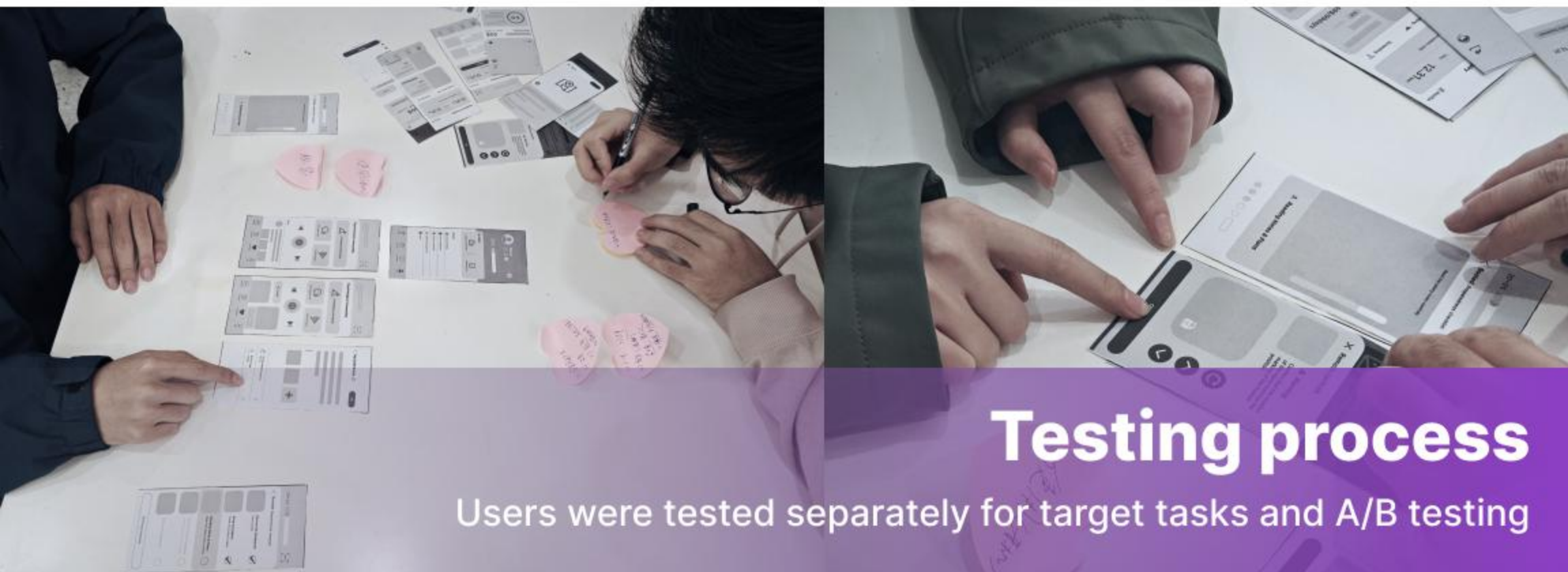
Demonstrate the design process from both the hardware product and the application software.

Logical plan



Brainstorming

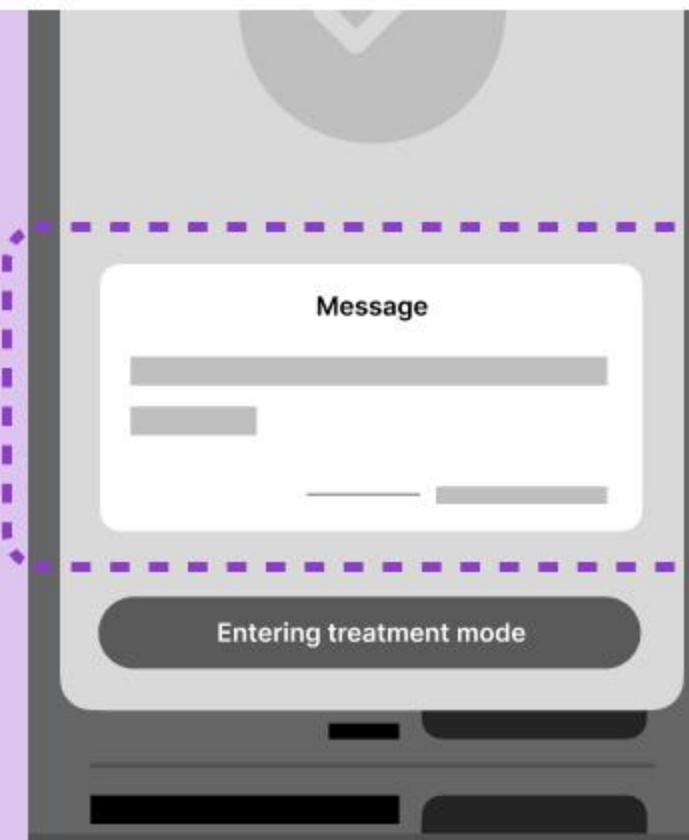




Bright spots

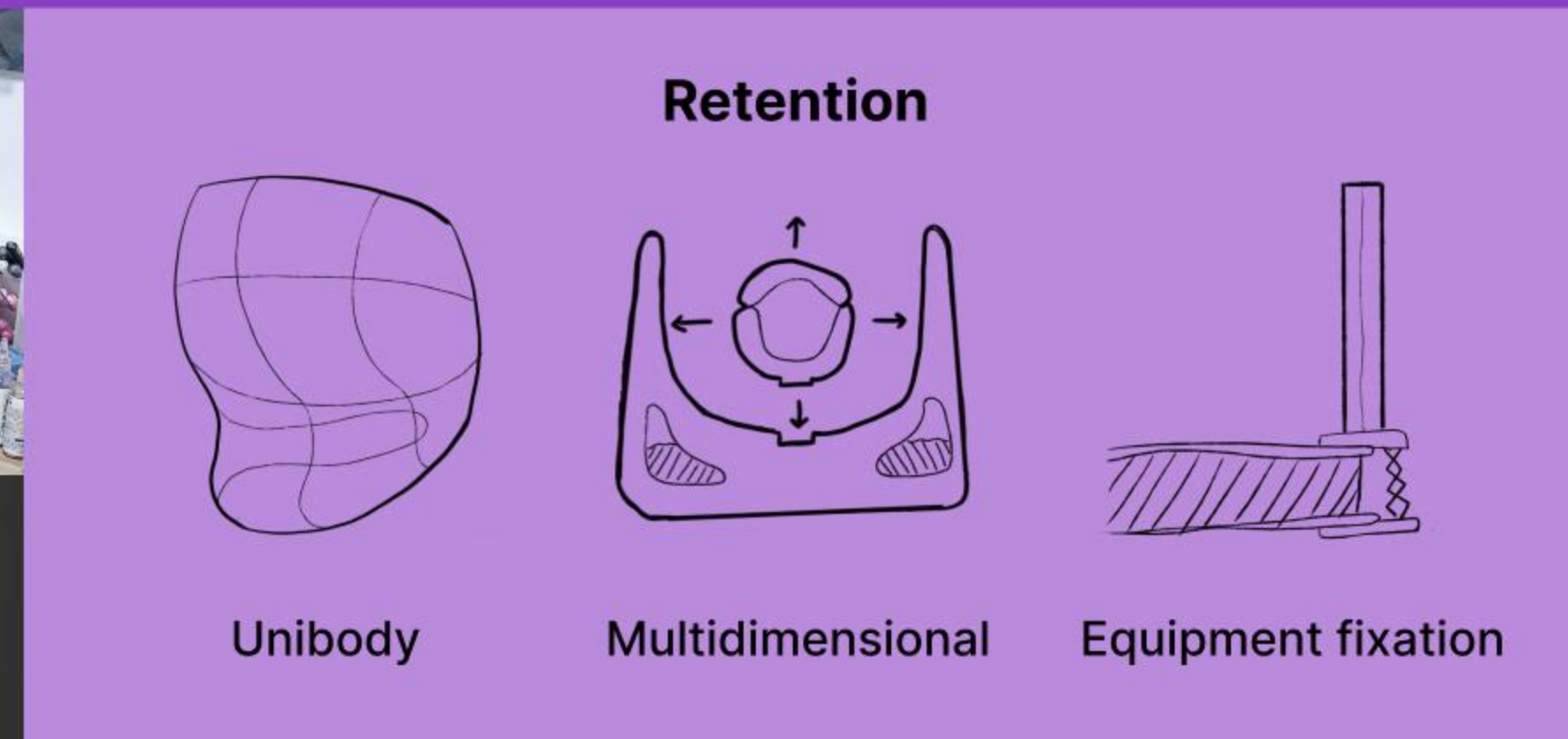
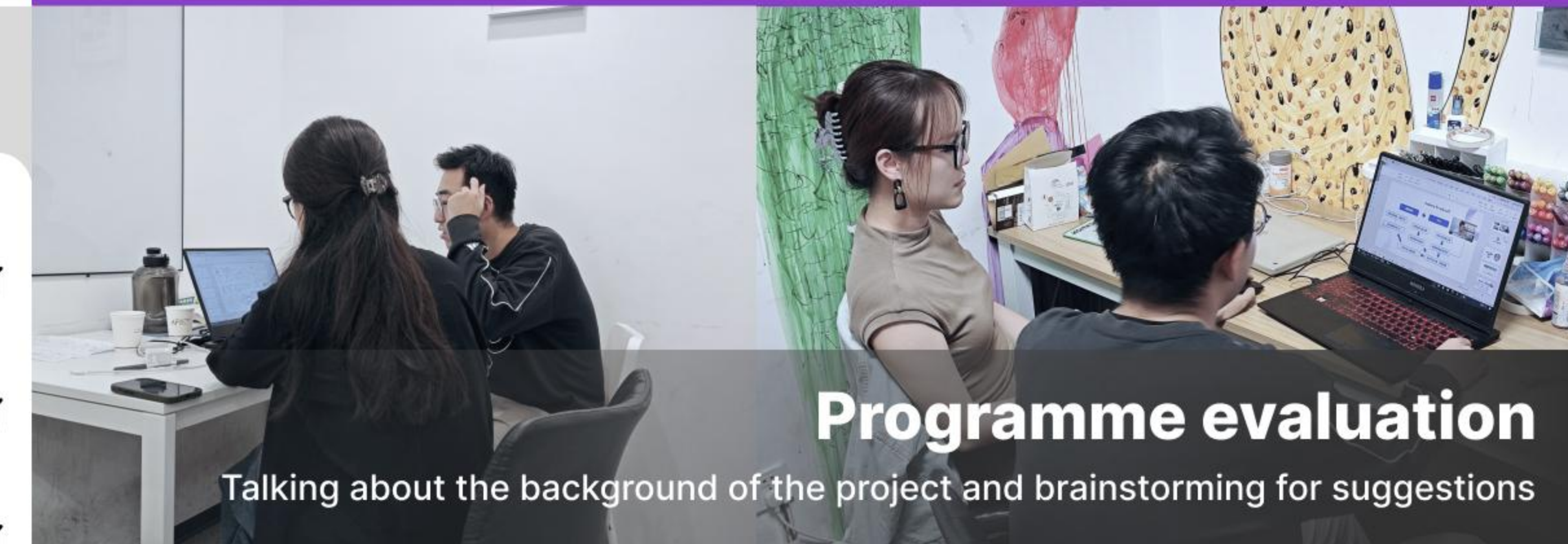
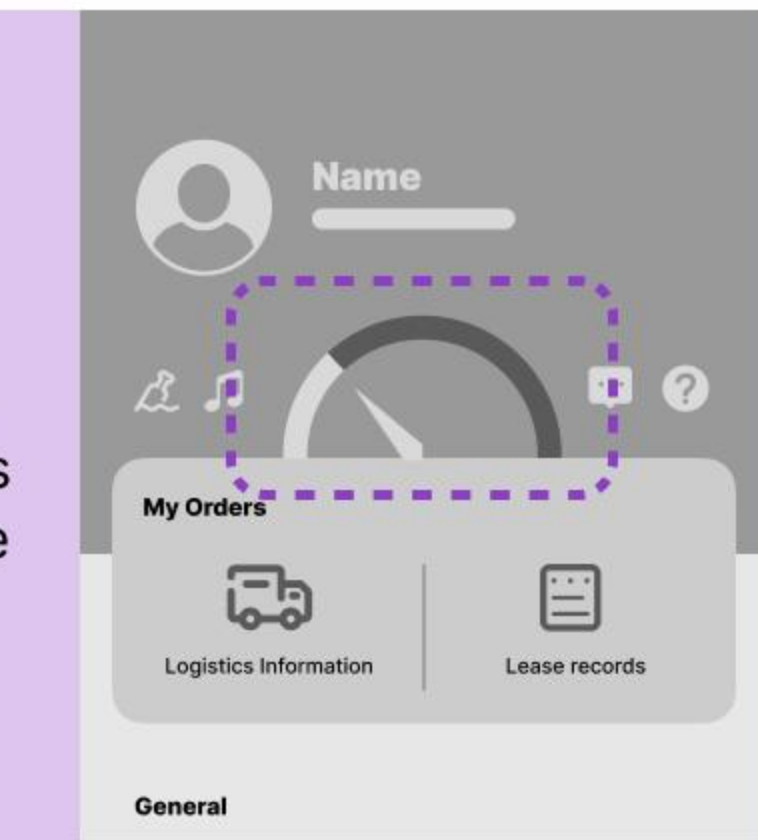
Societal

You will see the message left by the previous user when you bind for the first time.



Psychology

You can clearly see the current progress after the start of the treatment.

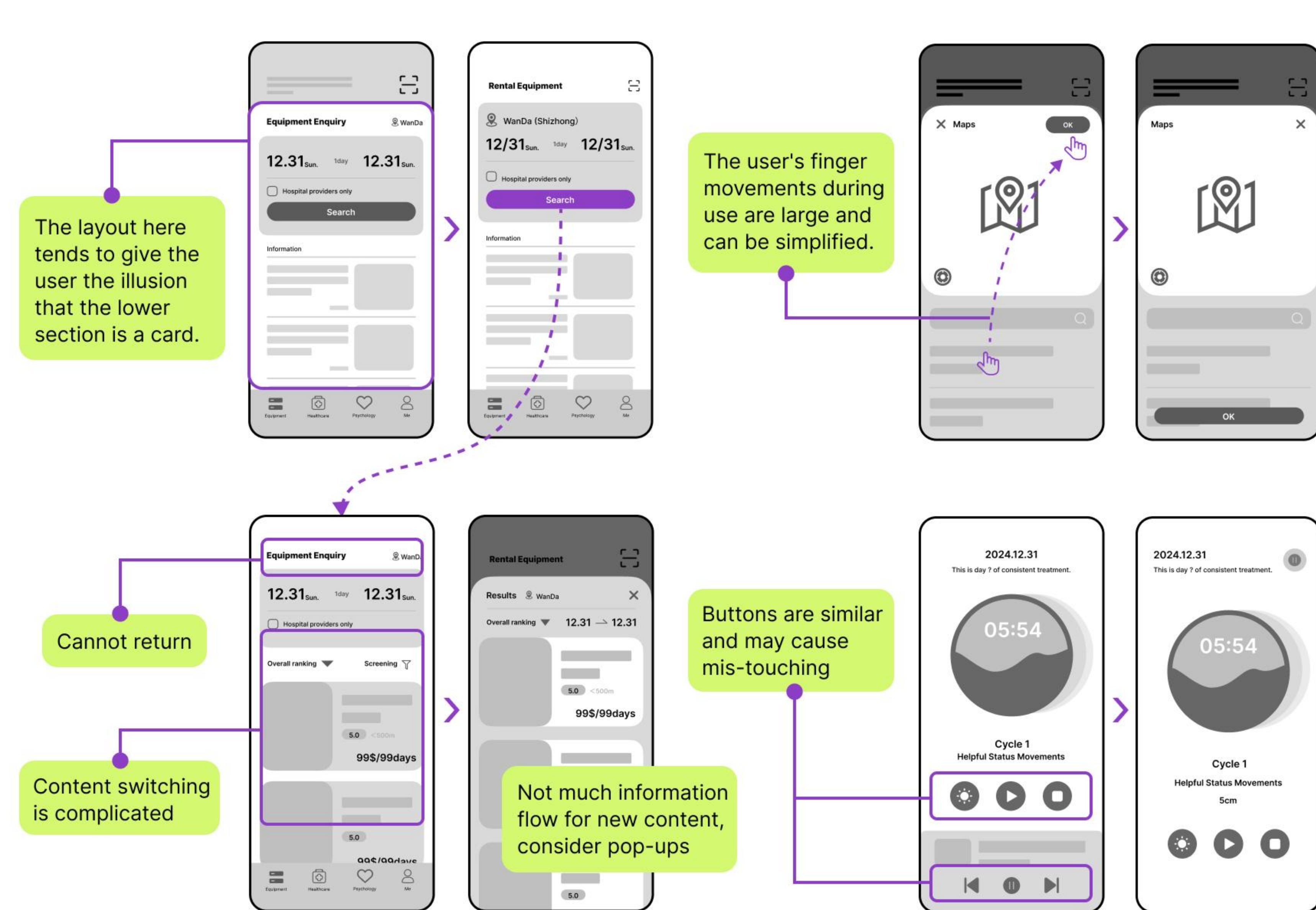


Suggestion

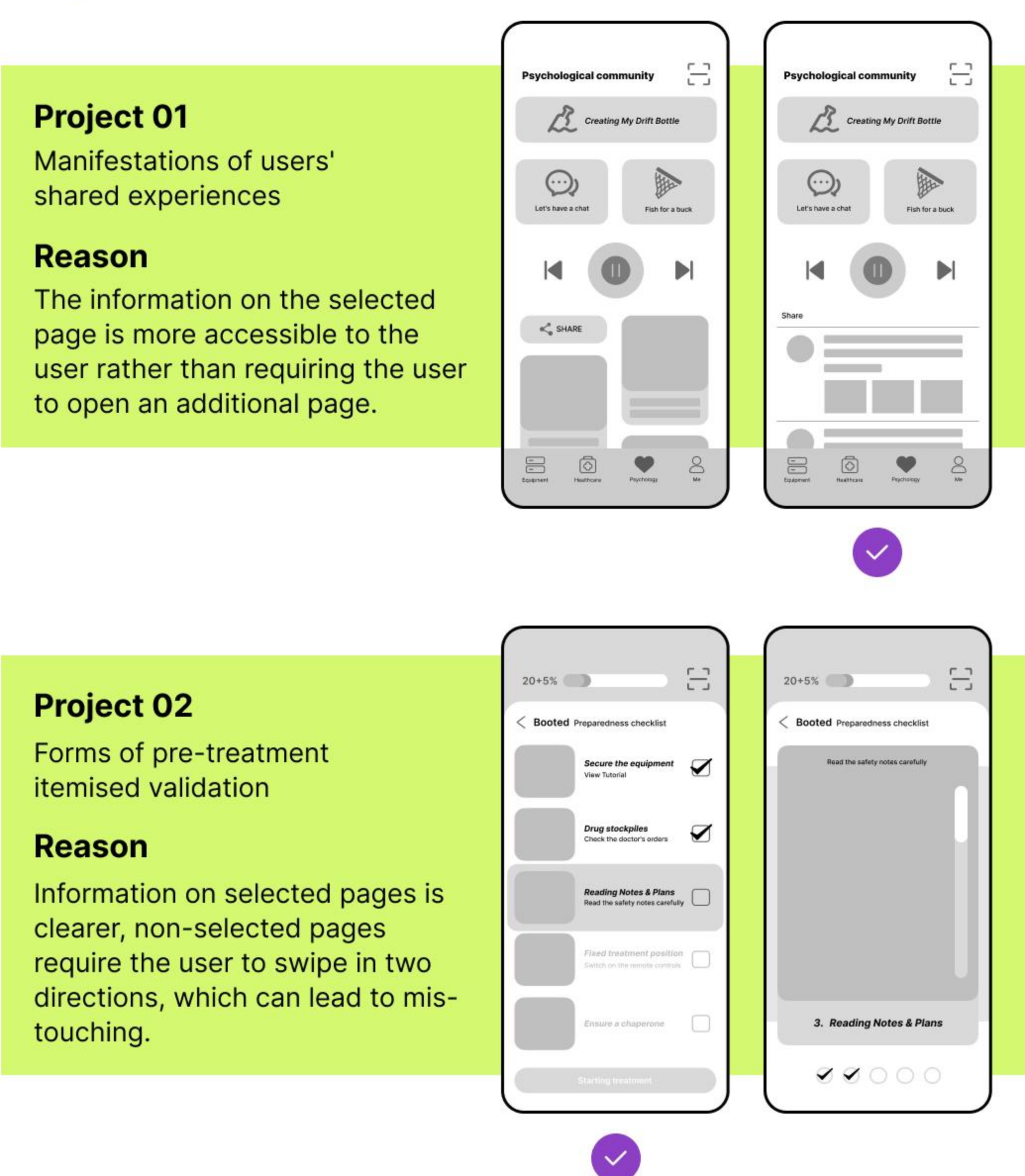
Rethinking physical structures that can be paired for better results.

Rethinking the form of fixing limb structures with reference to commercially available products

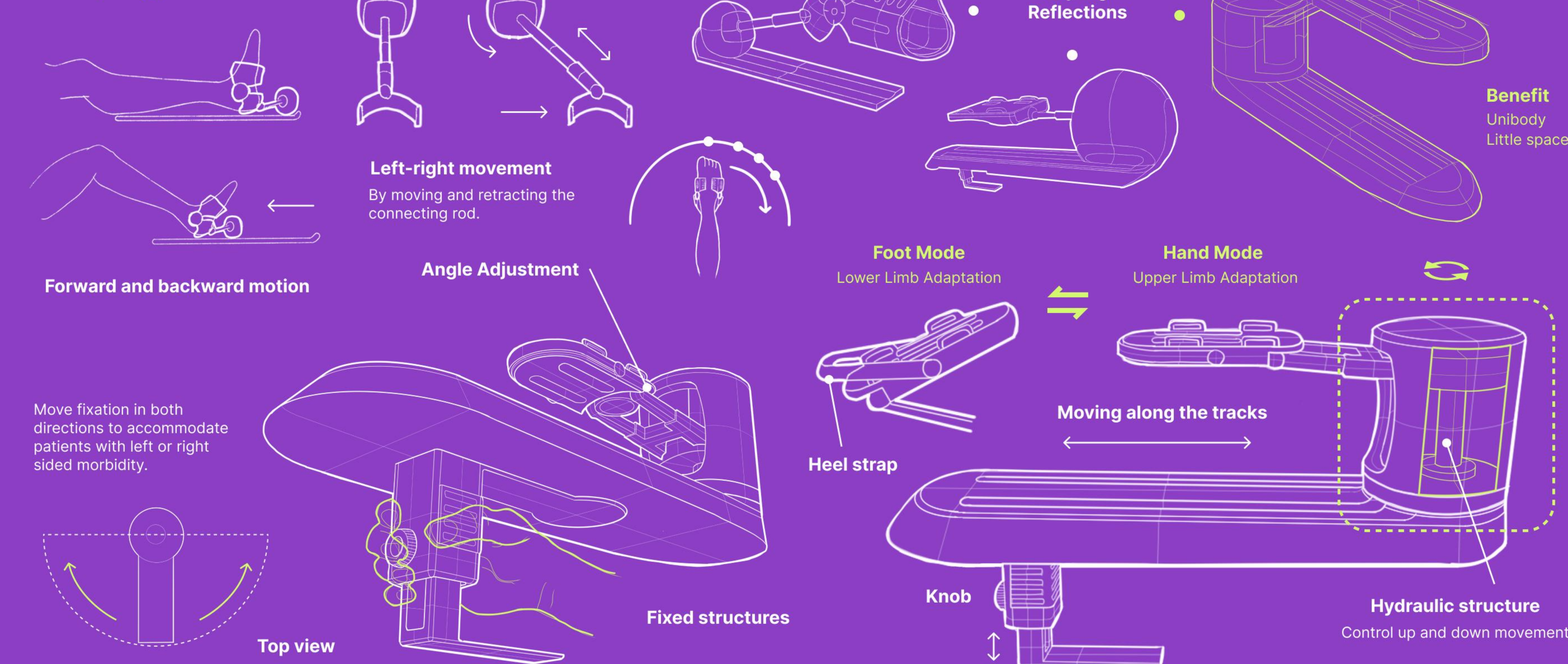
Interface modification



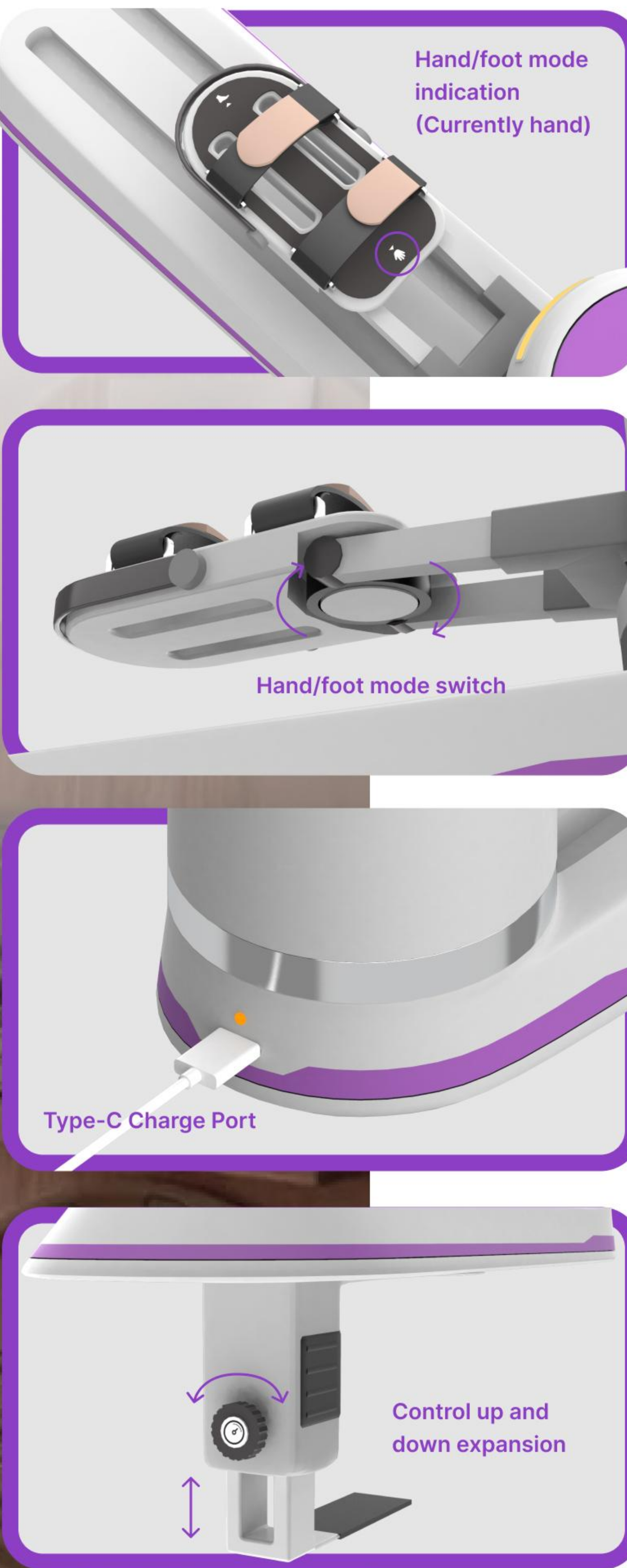
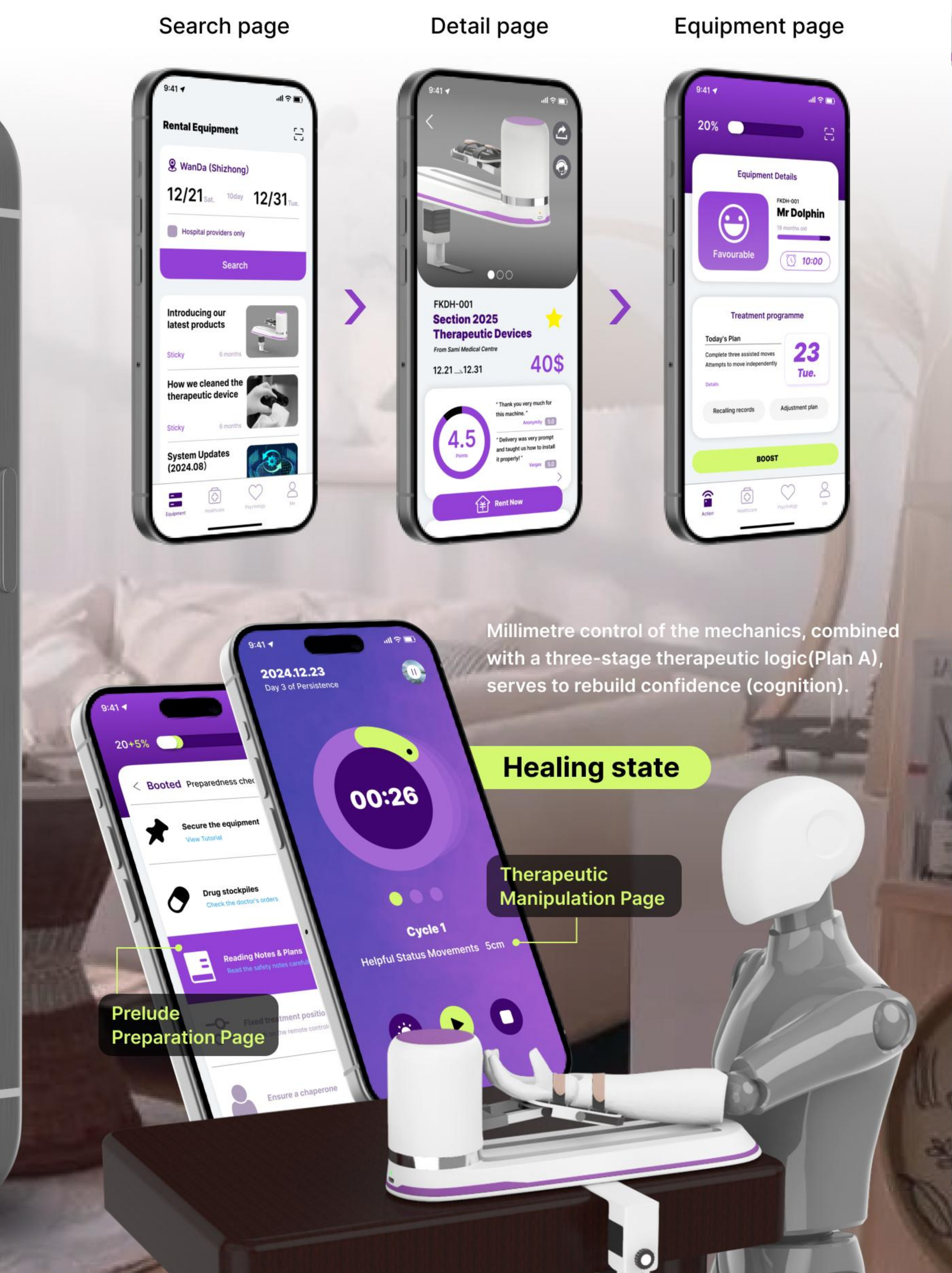
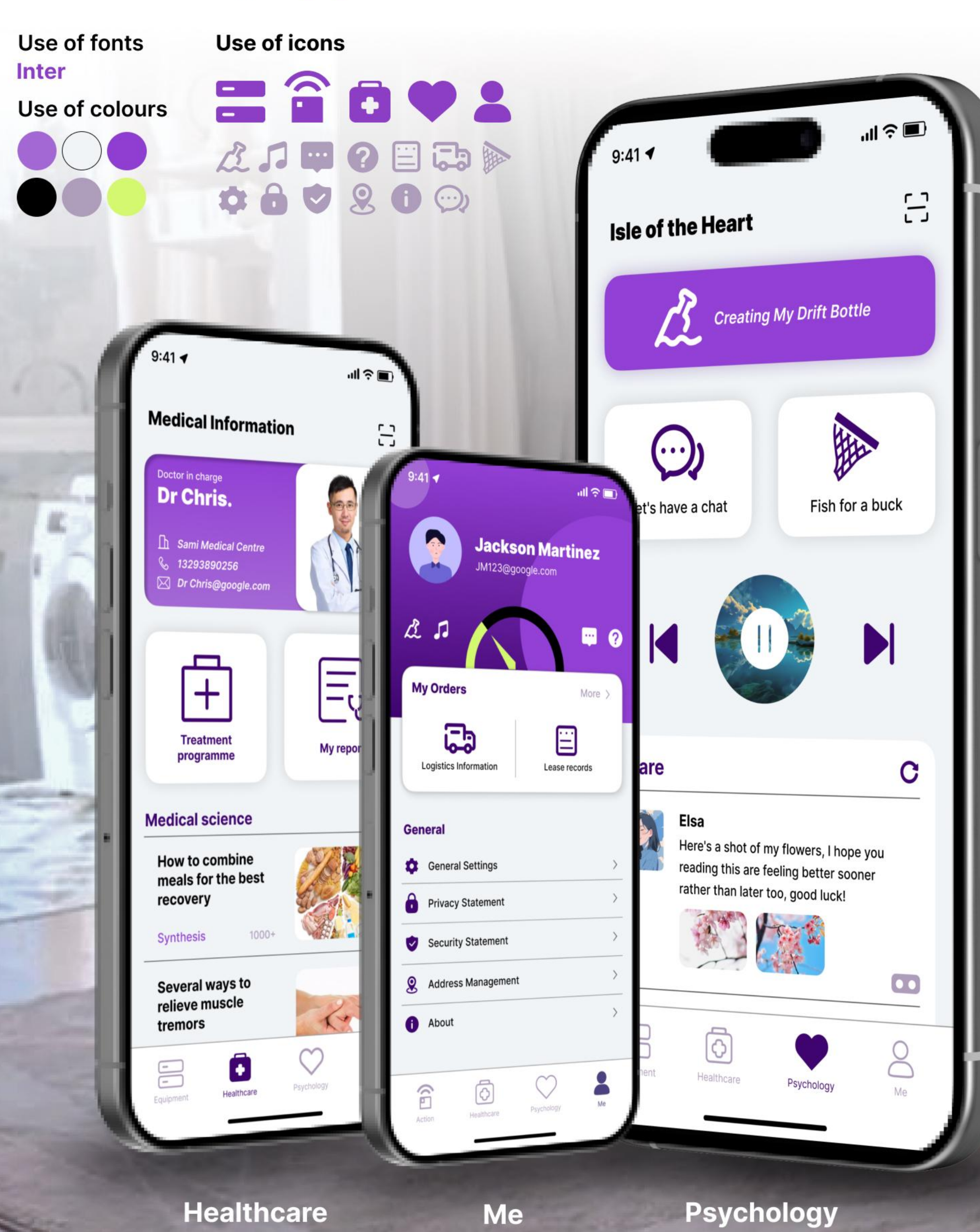
A/B Test



Final programme



Products/Applications Showcase



Exploded image

